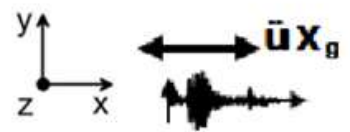
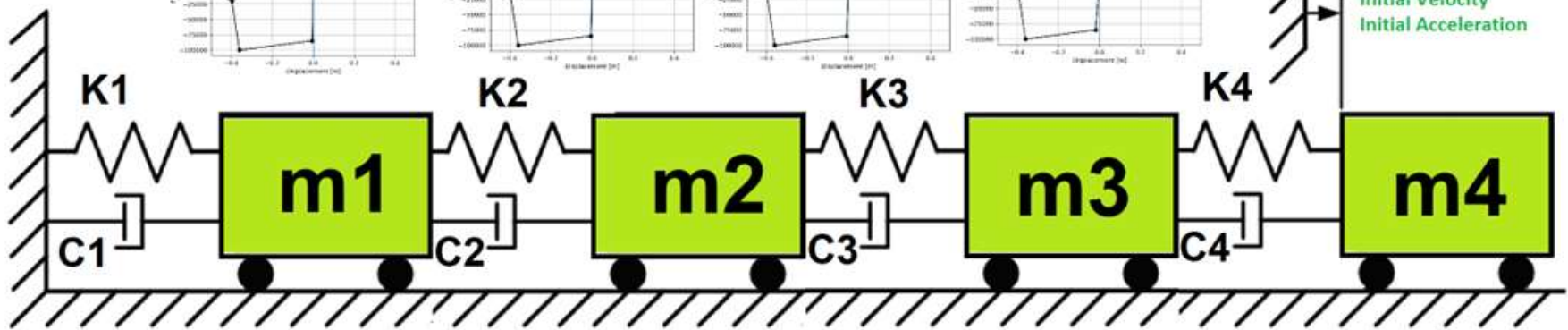
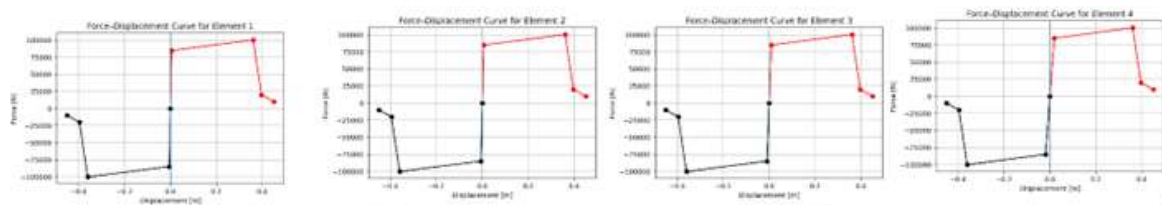


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION COMPUTATION AND VISUALIZATION RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OPENSEES

(CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM)

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

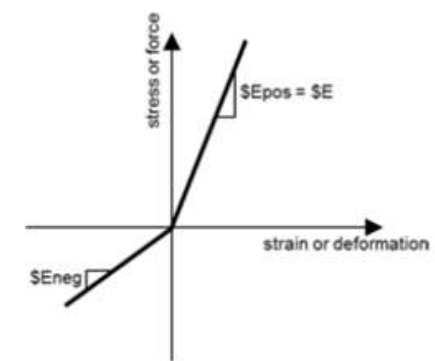
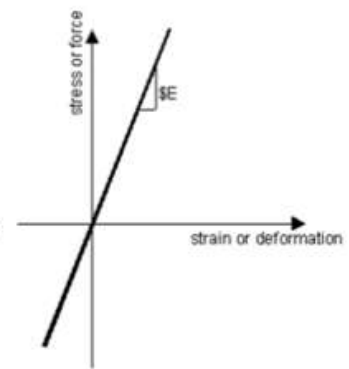


$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

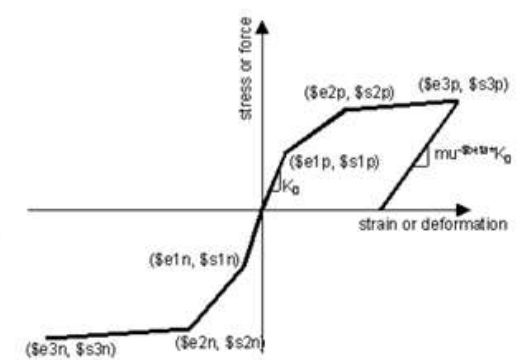
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

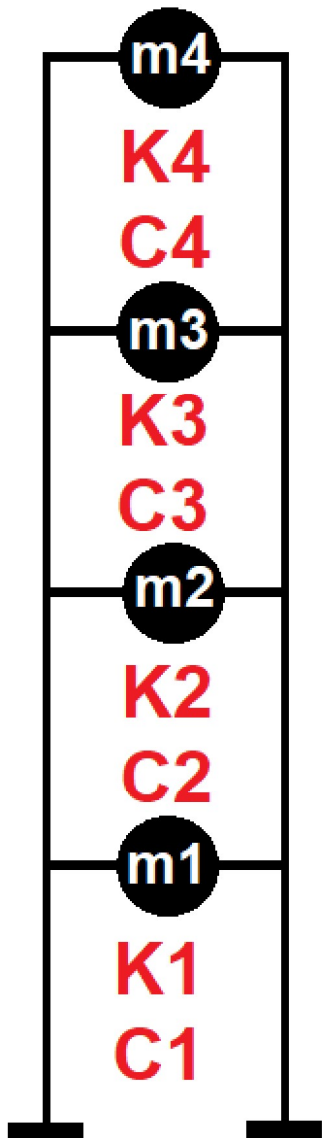
Δ_u = Lateral Ultimate Displaement from Pushover Analysis



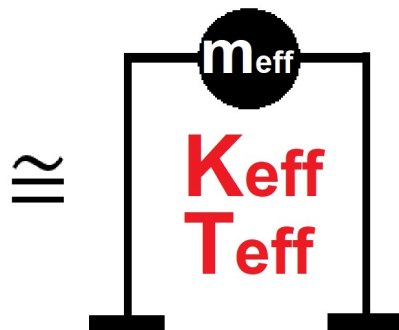
ELASTIC MATERIAL



INELASTIC MATERIAL

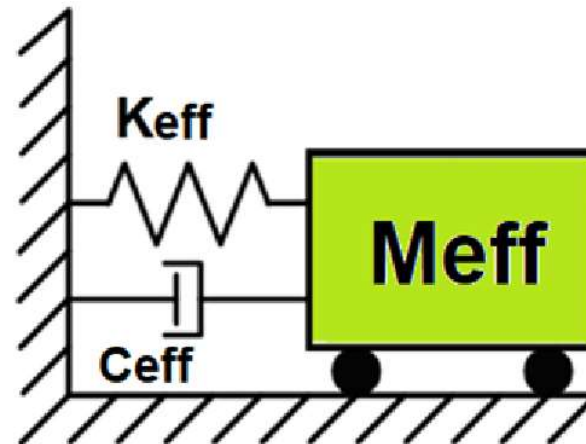


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

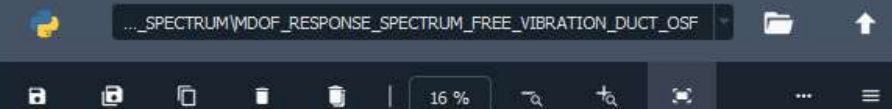
C:\Users\Dell\Desktop\OPENSEES_FILES\CONSTANT_STR...MDOF_RESPONSE_SPECTRUM_FREE_VIBRATION_DUCT_OSF.py

MDOF_RESPONSE_SPEC...RATION_DUCT_OSF.py X

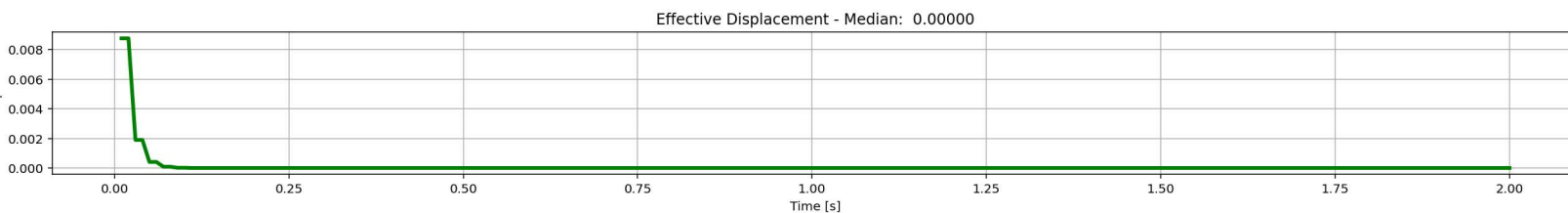
```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION AND VISUALIZATION
4 # RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING C
5 #
6 # INVESTIGATION OF THE EFFECTS OF MDOF STRUCTURAL DUCTILITY ON DAMAGE LEVEL AND STRUCTURAL
7 # USING NONLINEAR DYNAMIC ANALYSIS
8 #
9 #
10 #-----
11 # CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM
12 #-----
13 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH FREE
14 #
15 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
16 #-----
17 # IT IS RECOMMENDED FOR BETTER SEEING THE STRUCTURAL DUCTILITY AND STRUCTURAL OVER-STRENGTH F
18 # INITIAL DISPLACEMENT HAVE TO GREATER THAN YIELD DISPLACEMENT.
19 #-----
20 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
21 # EMAIL: salar.d.ghashghaei@gmail.com
22 #####
23 """
24 Investigation of the Effects of Structural Ductility on Damage Level and Structural Damping:
25 This code implements a comprehensive nonlinear dynamic analysis framework for
26 performance-based earthquake engineering assessment of single-degree-of-freedom
27 (SDOF) systems. The methodology combines traditional nonlinear time-history
28 analysis with modern probabilistic and machine learning techniques for advanced
29 structural performance evaluation with changing structural ductility ratio and over strength f
30
31 KEY ENGINEERING OBJECTIVES:
32 1. Comparative assessment of hysteretic models for free-vibration response prediction
33 2. Probabilistic free-vibration demand analysis using multiple ground motions
34 3. Development of fragility curves for performance-based earthquake engineering

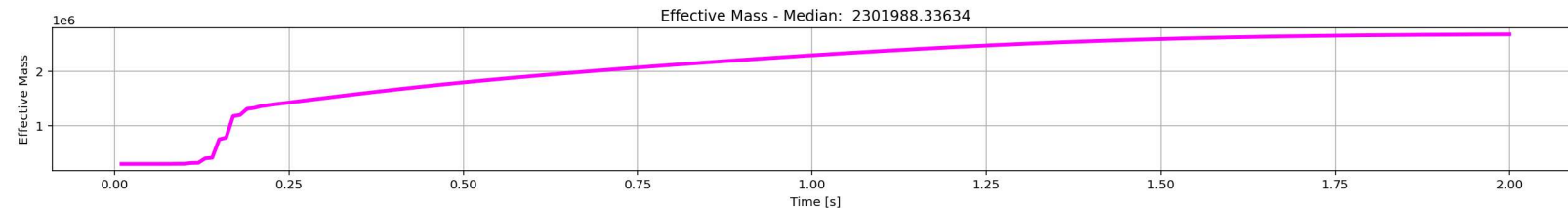
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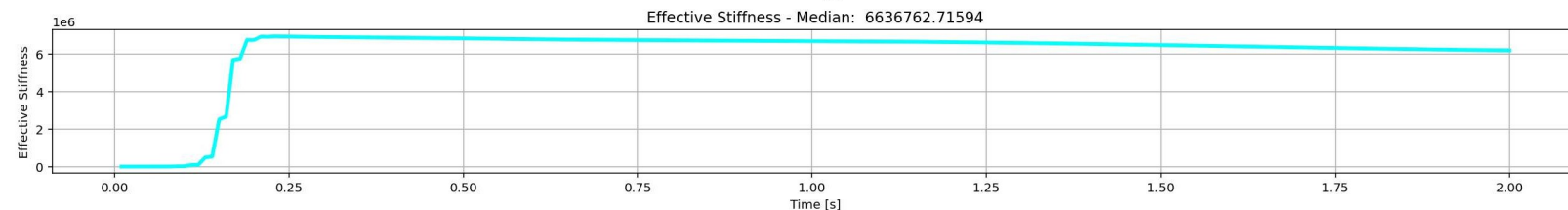
[Python Console] Plots Variable Explorer Help Debugger History Files



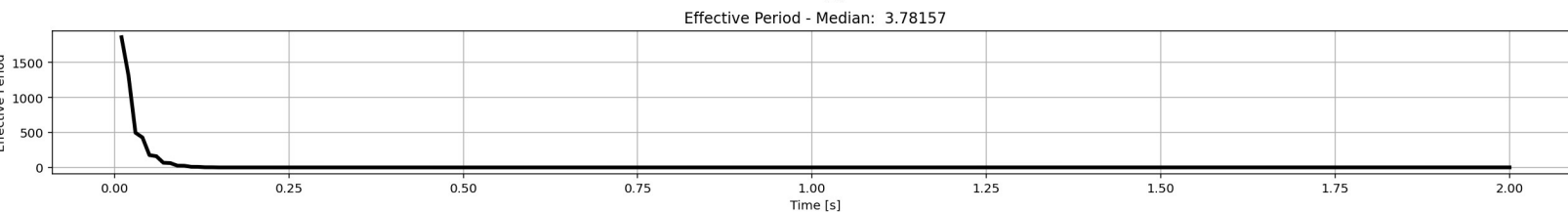
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



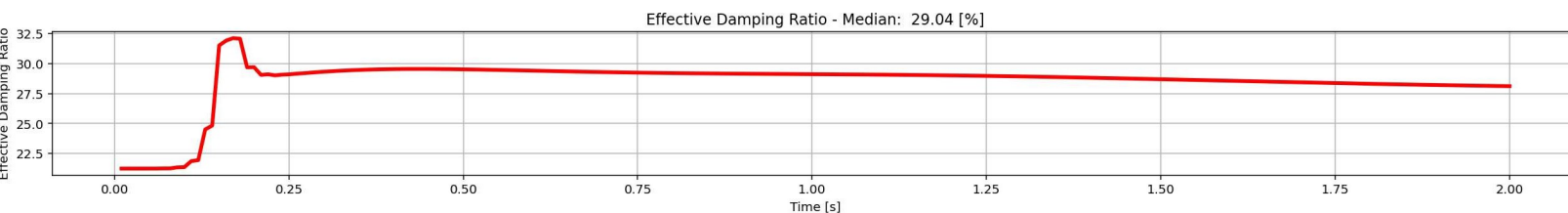
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



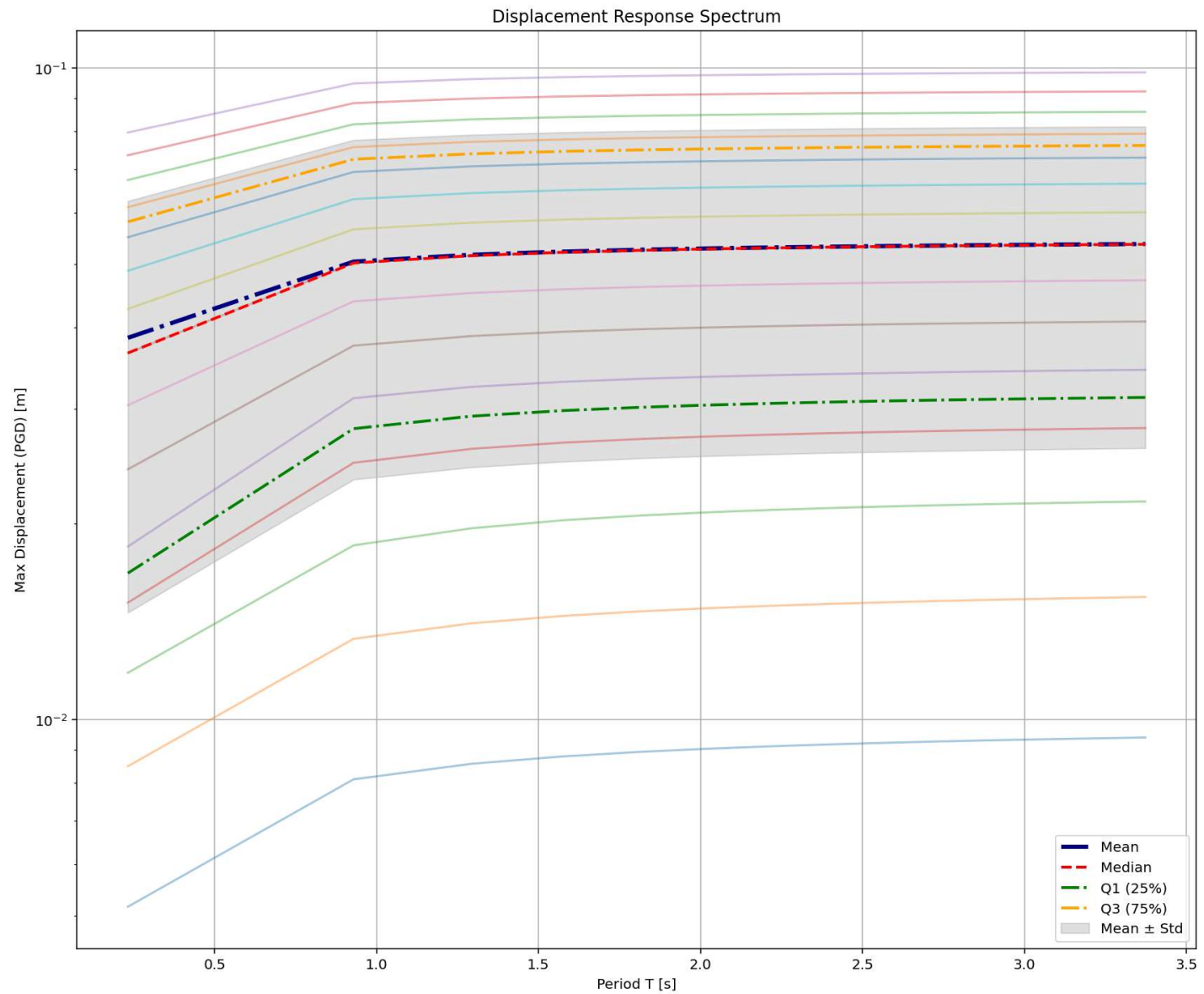
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

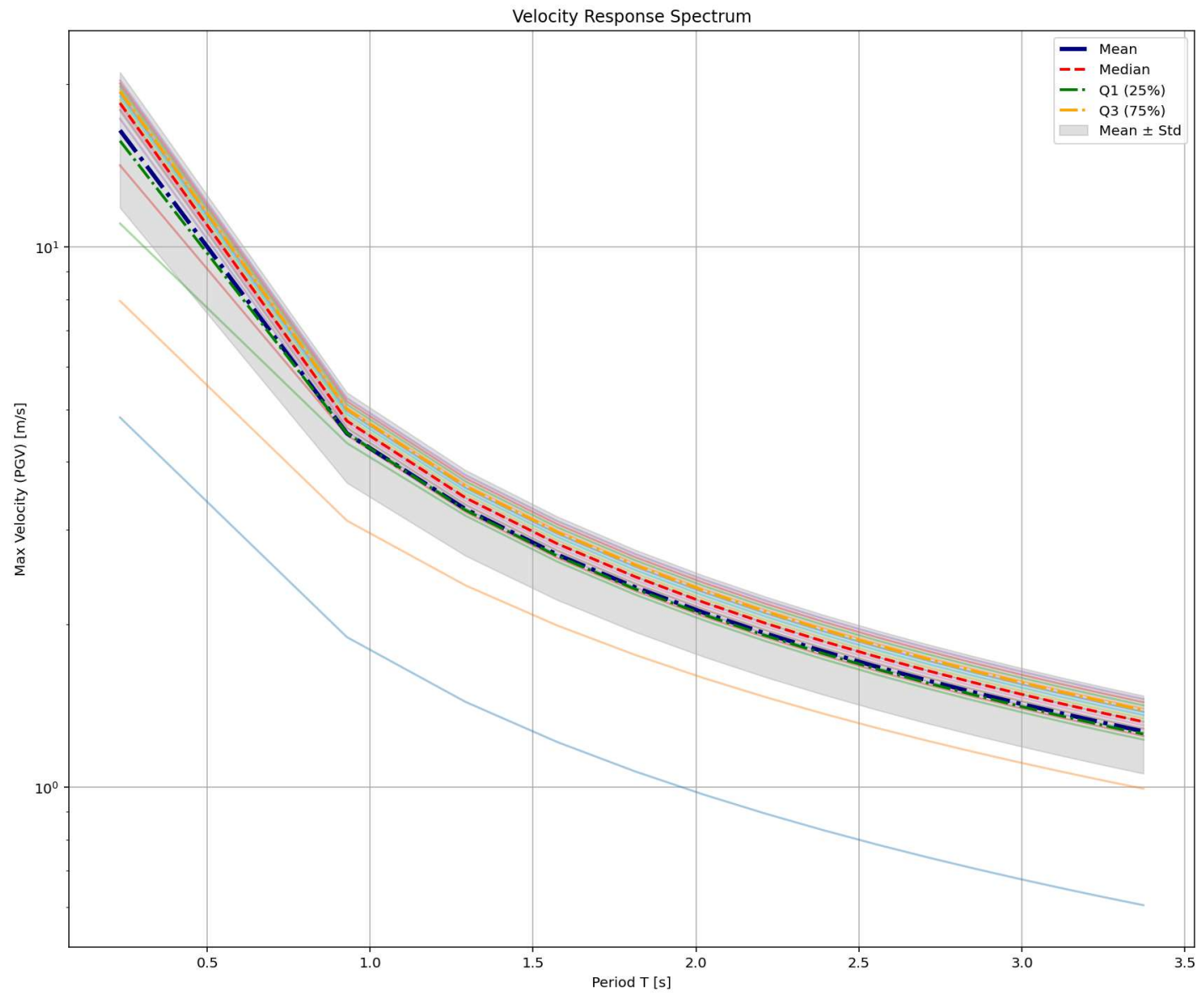


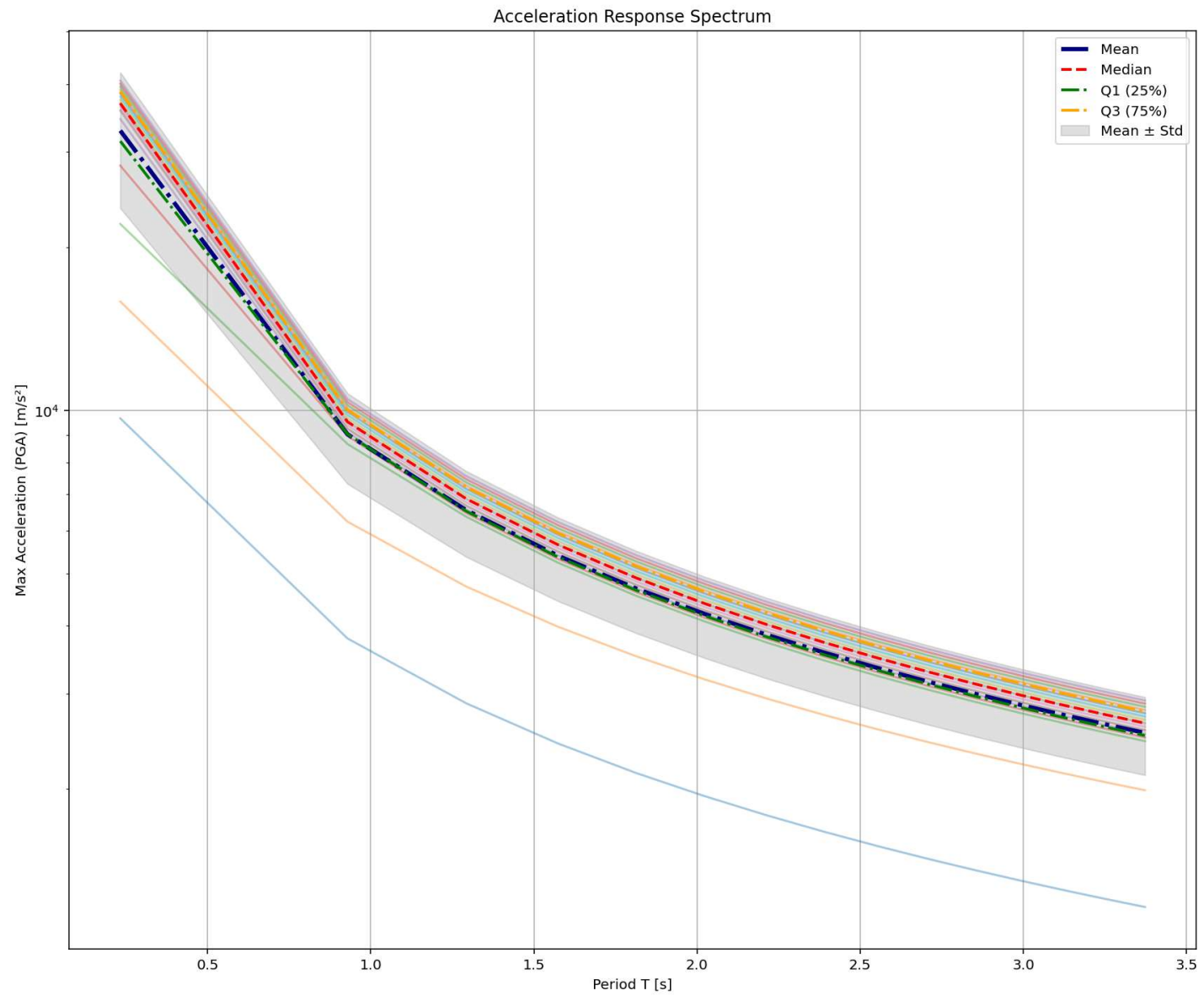
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

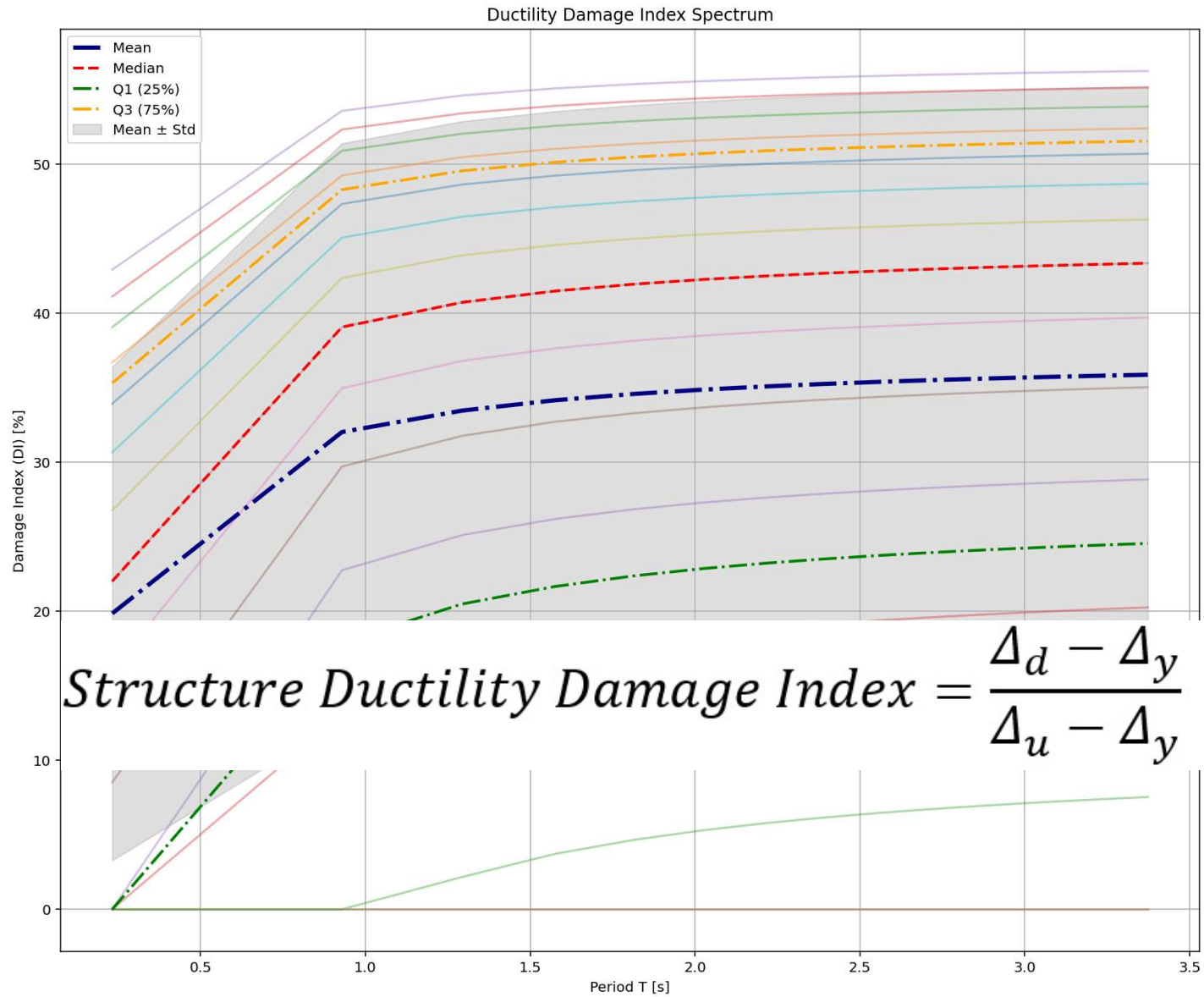


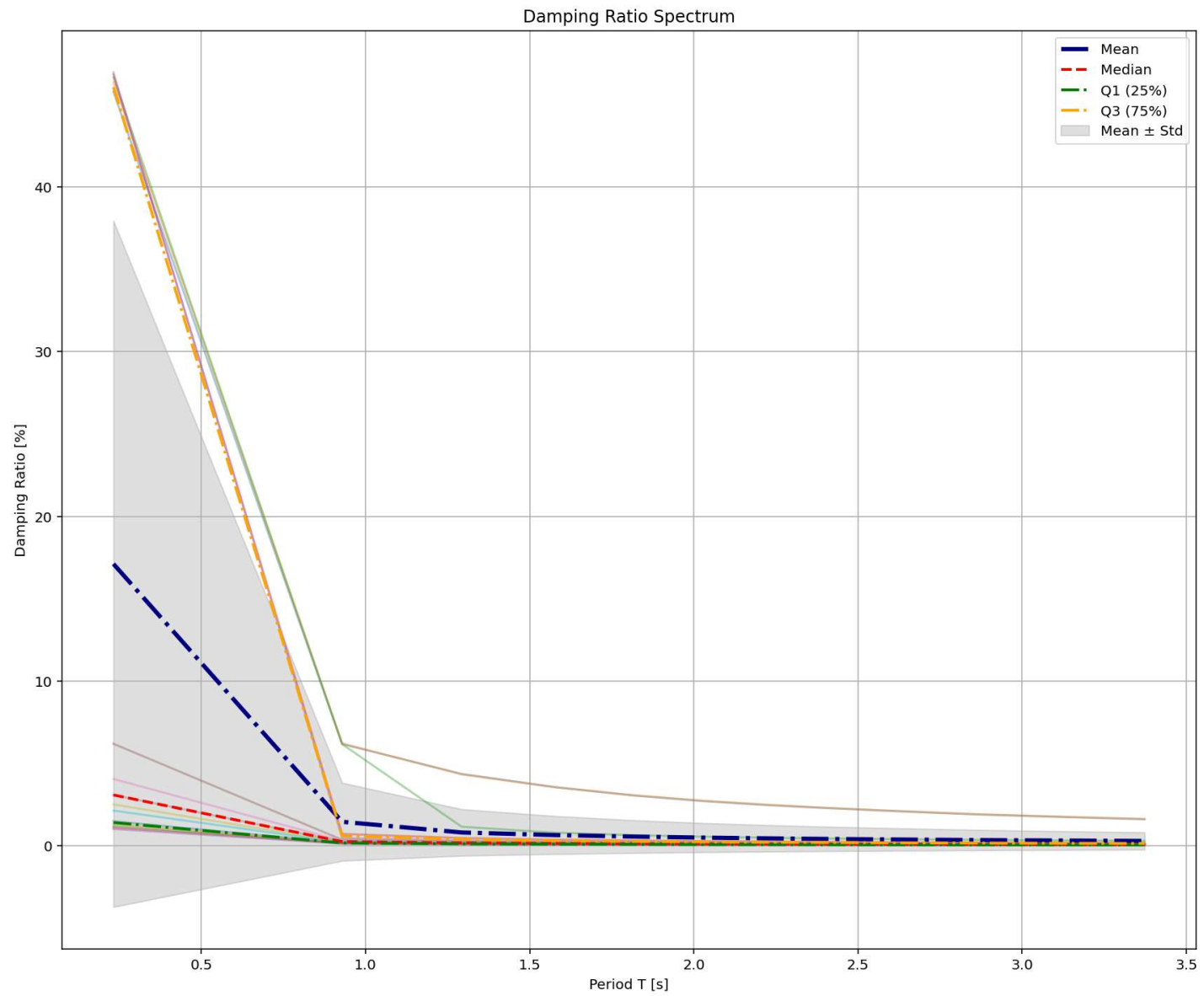
$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$

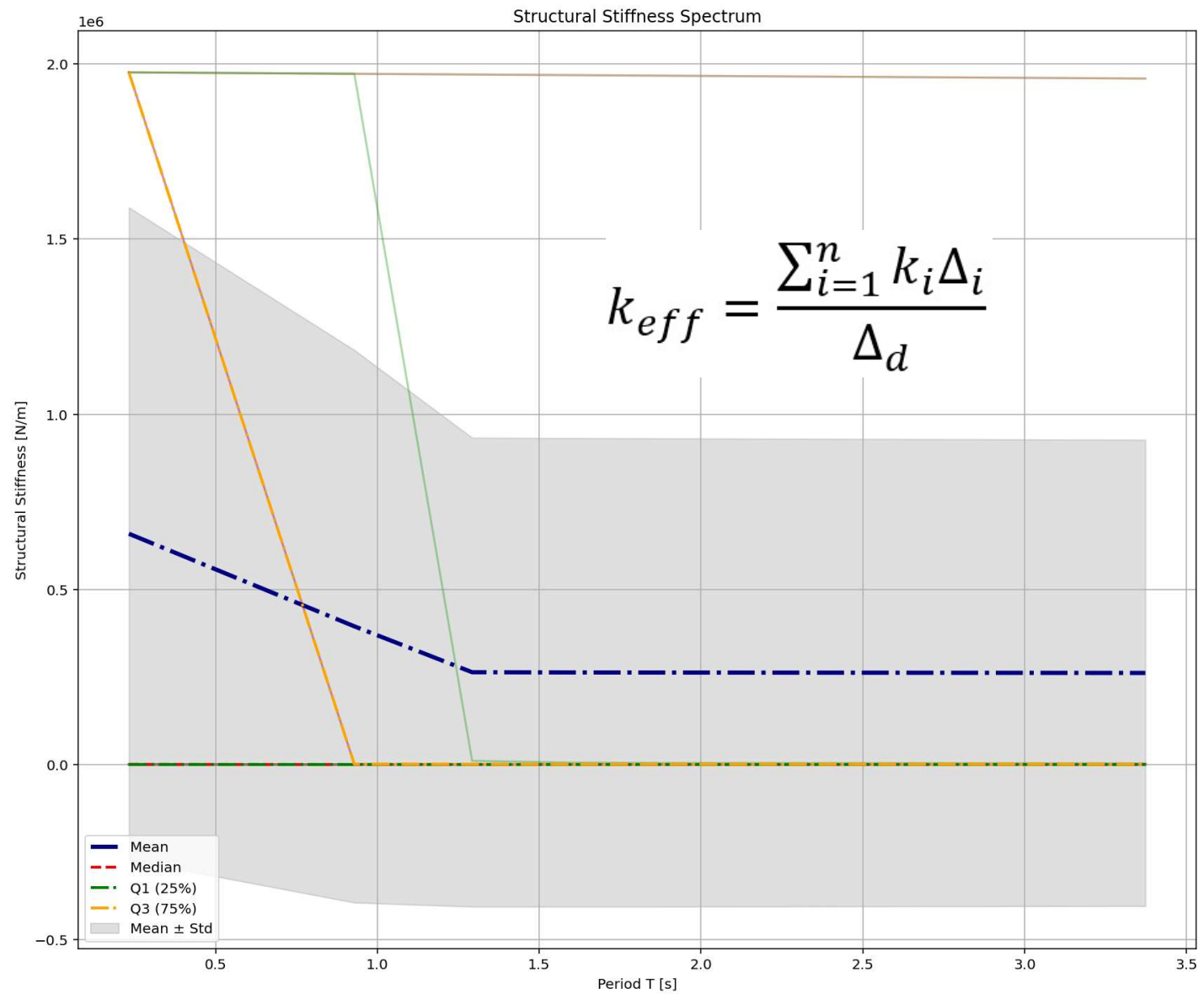


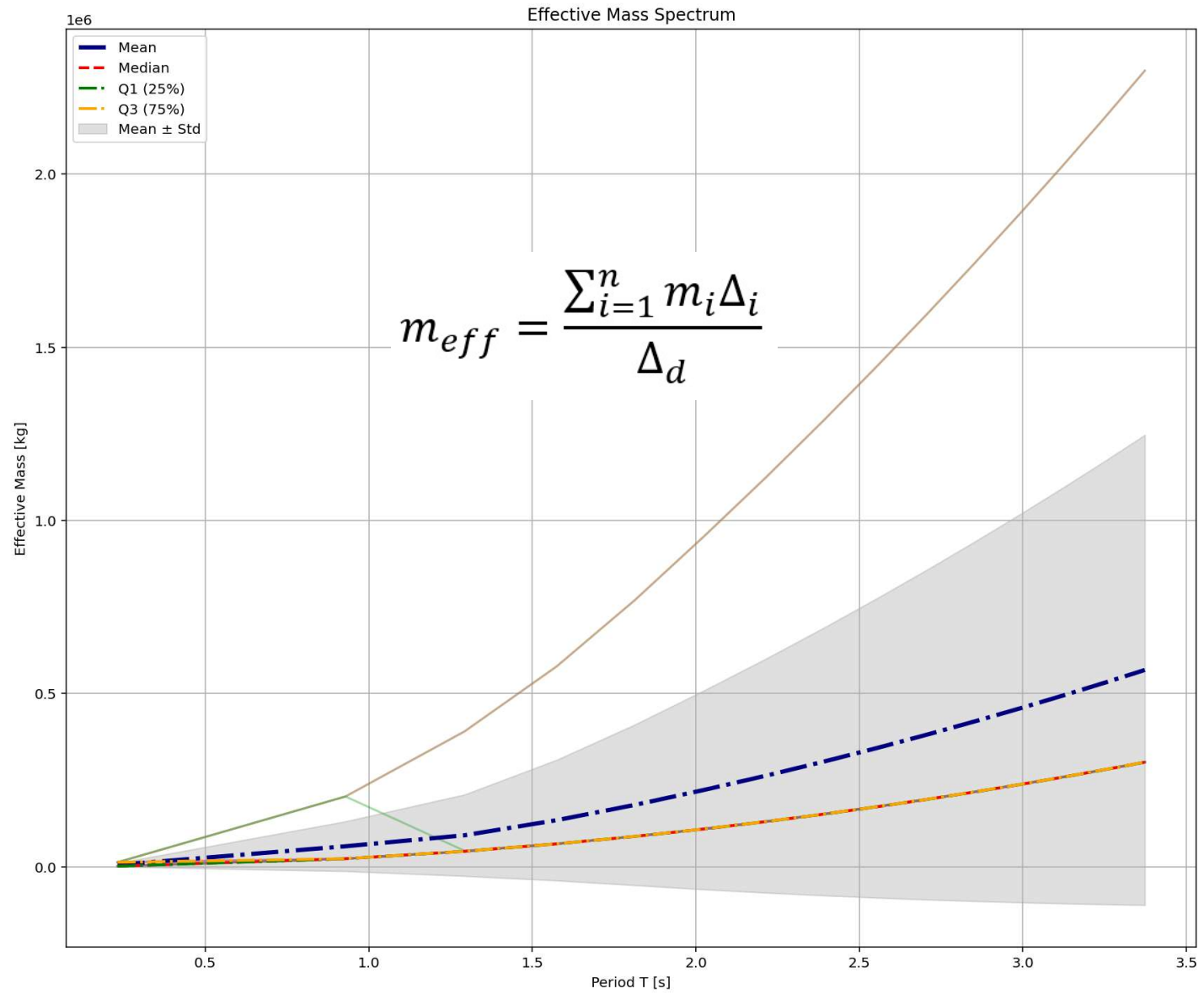


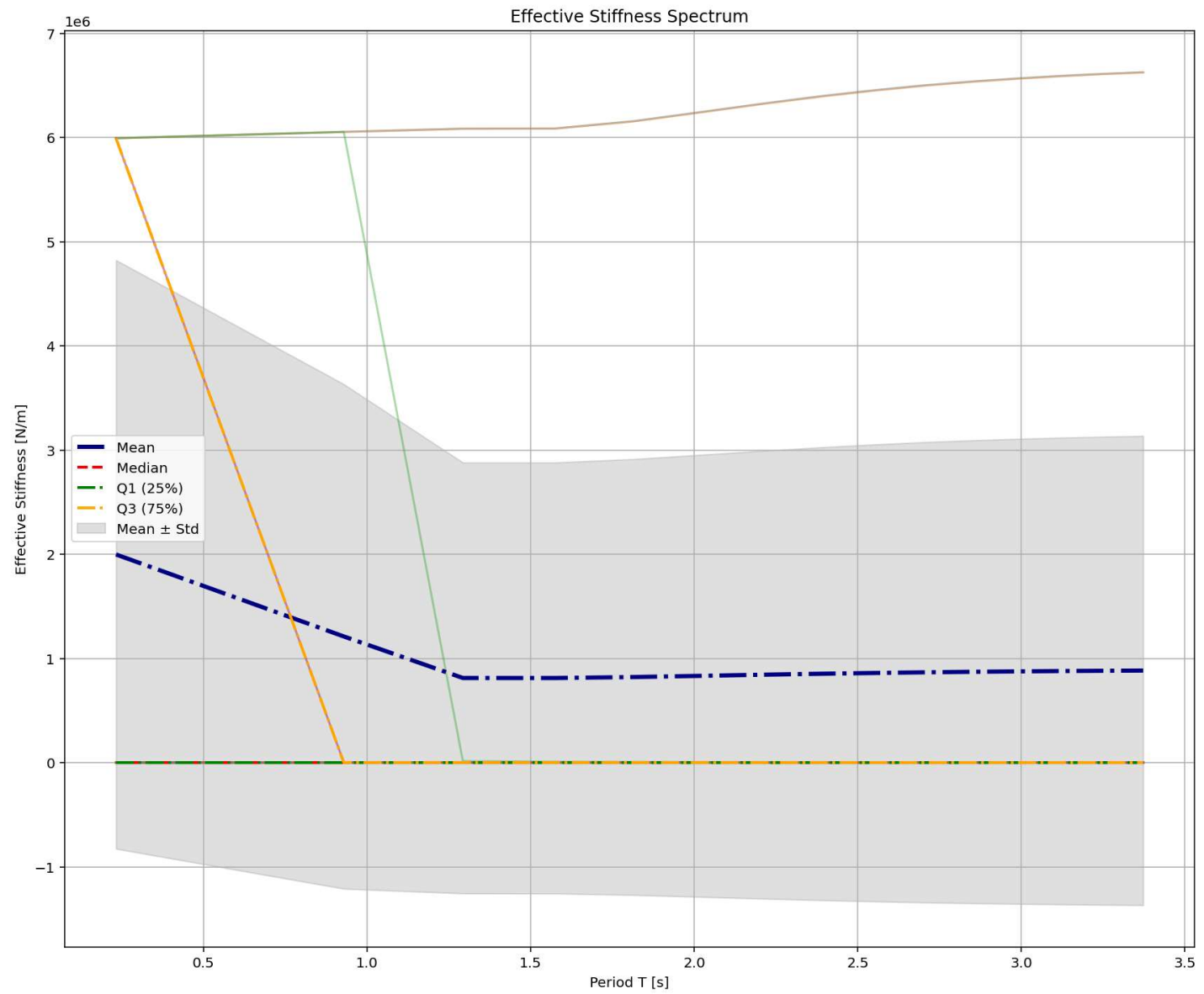


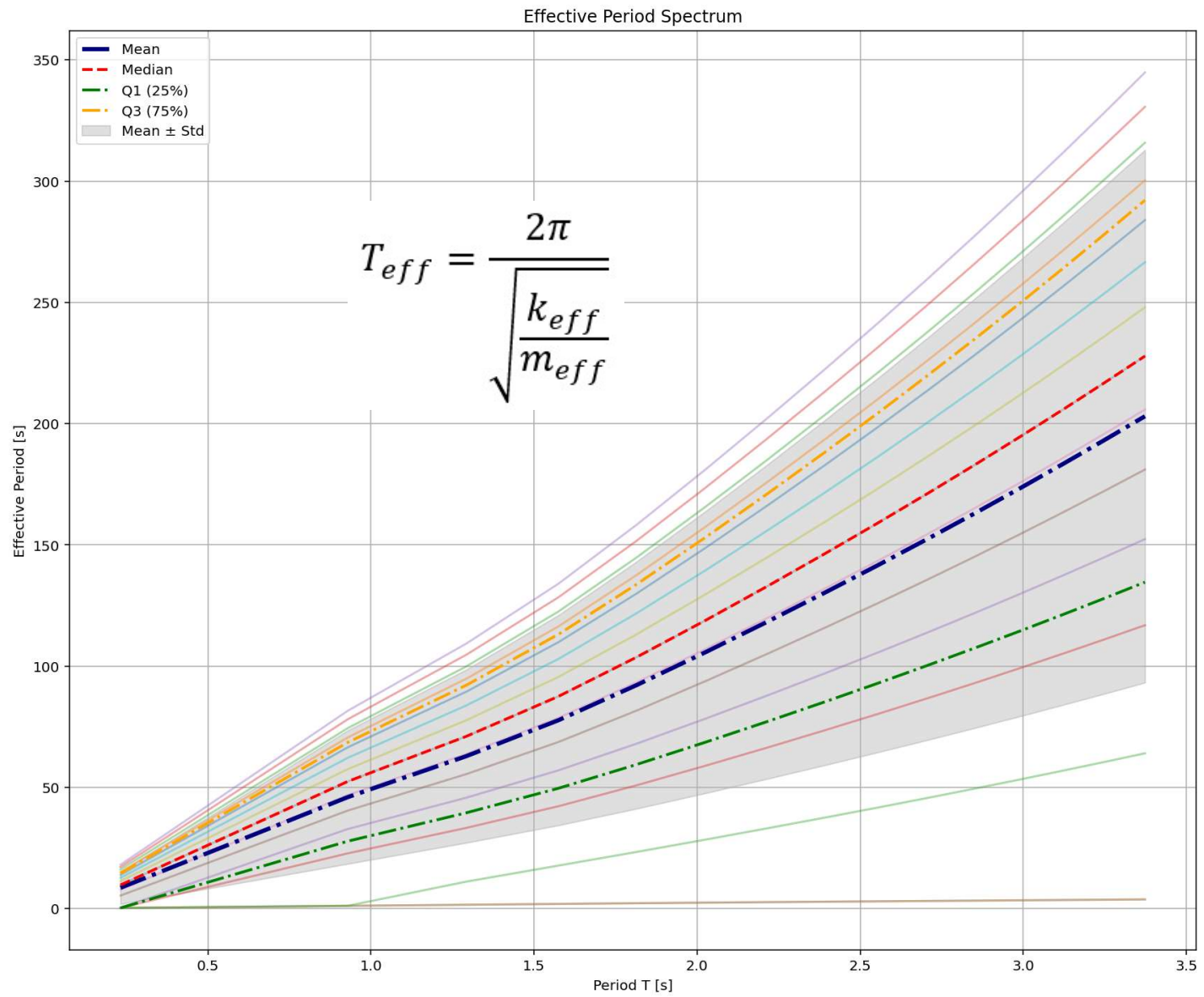


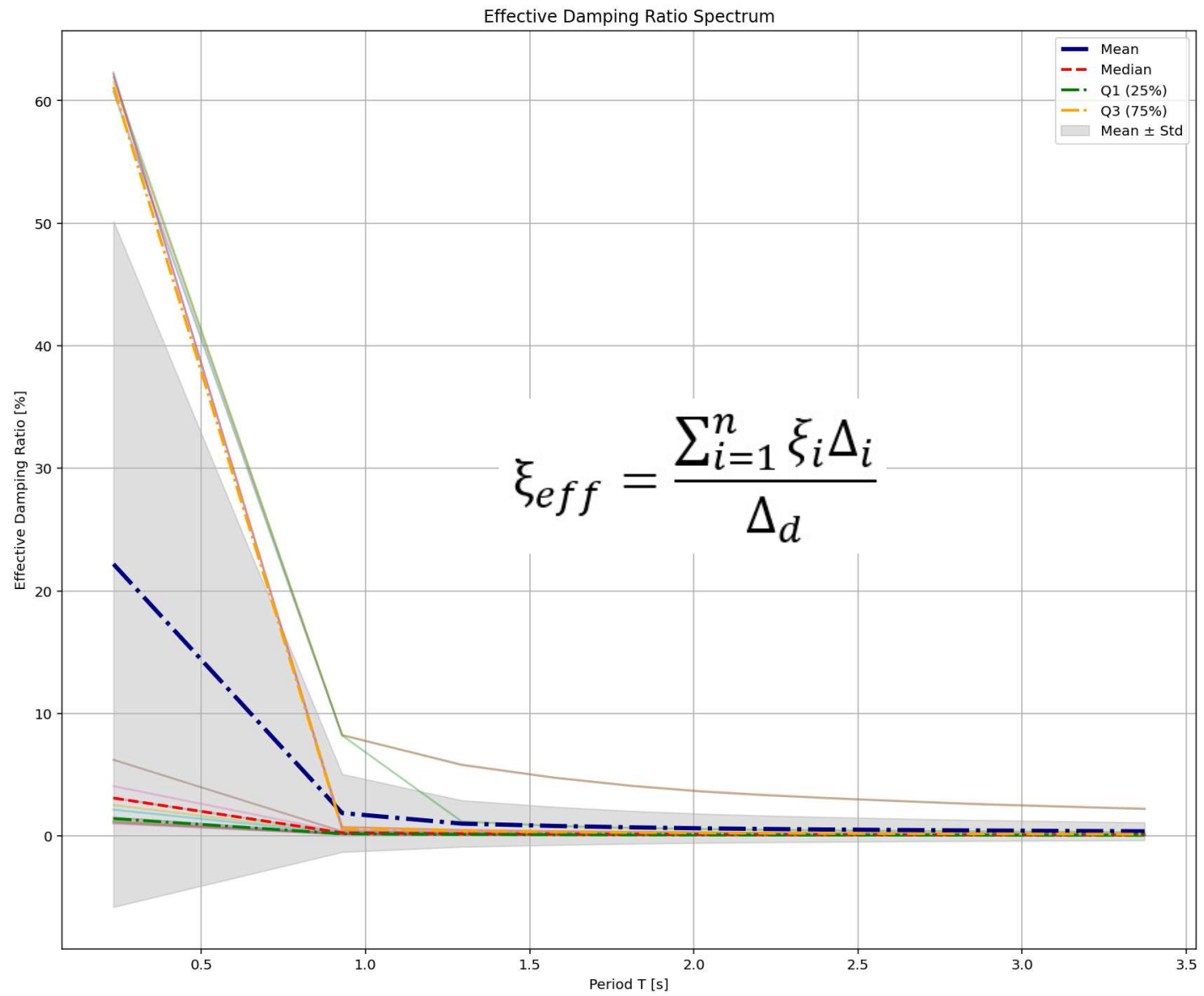


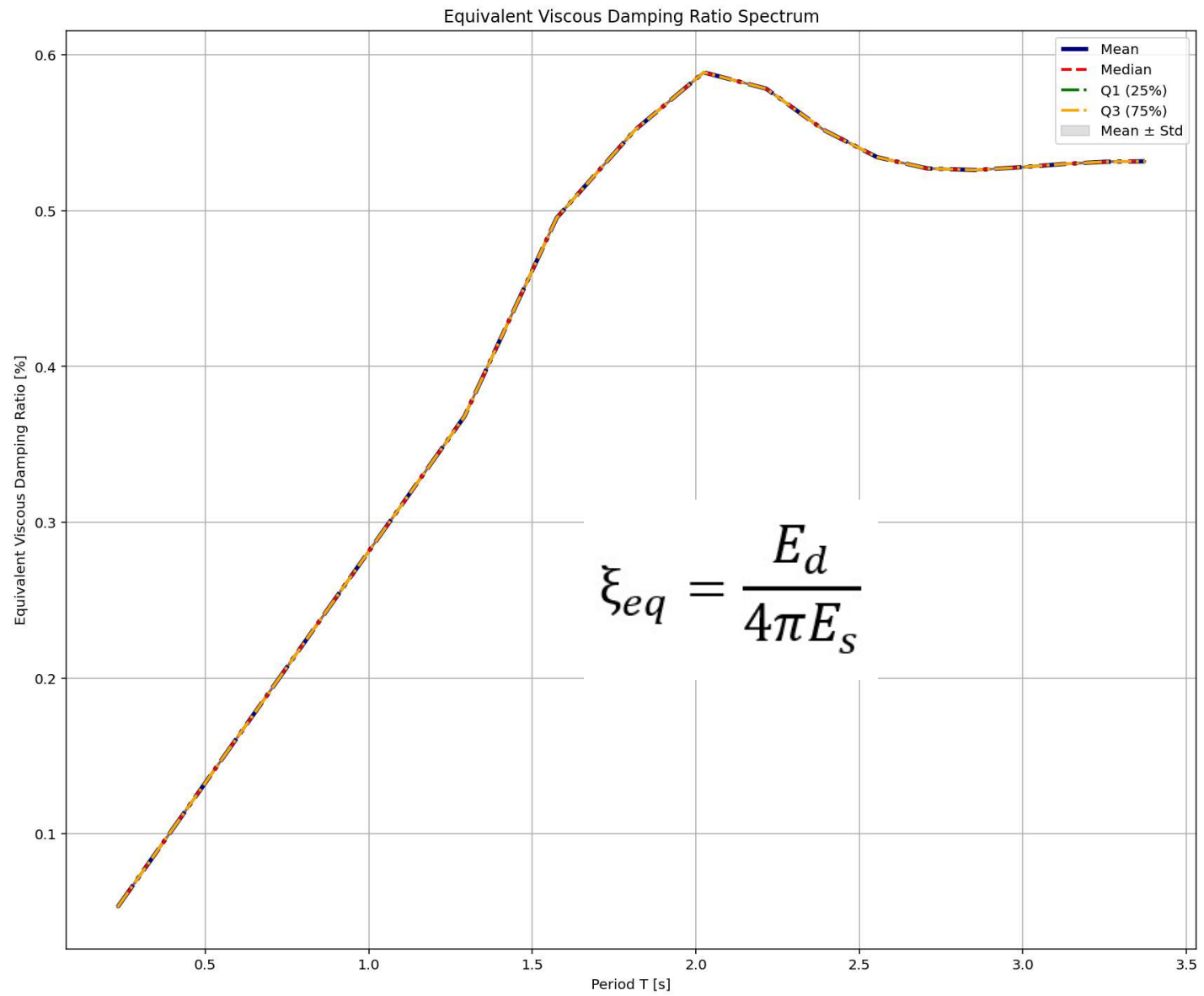


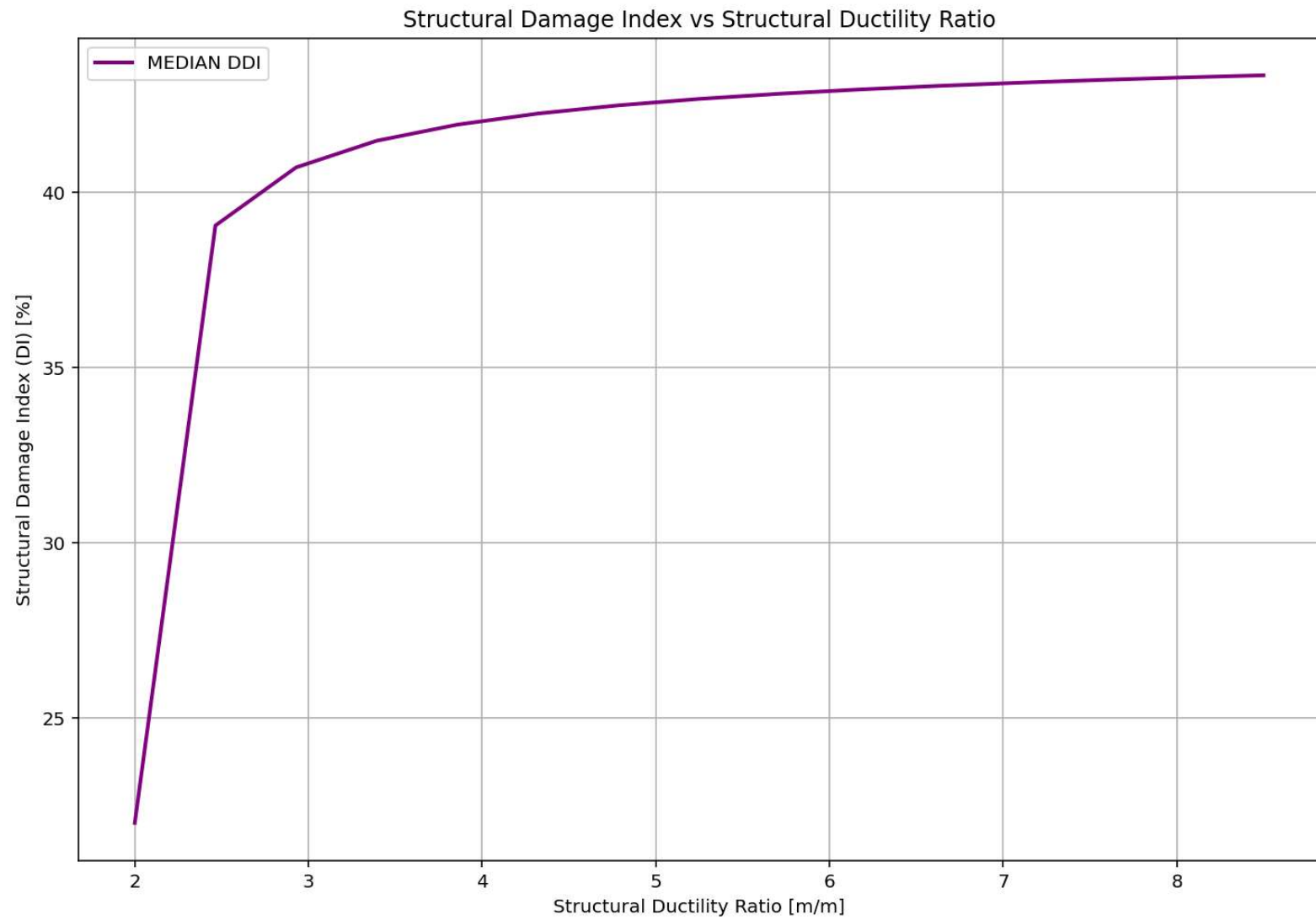


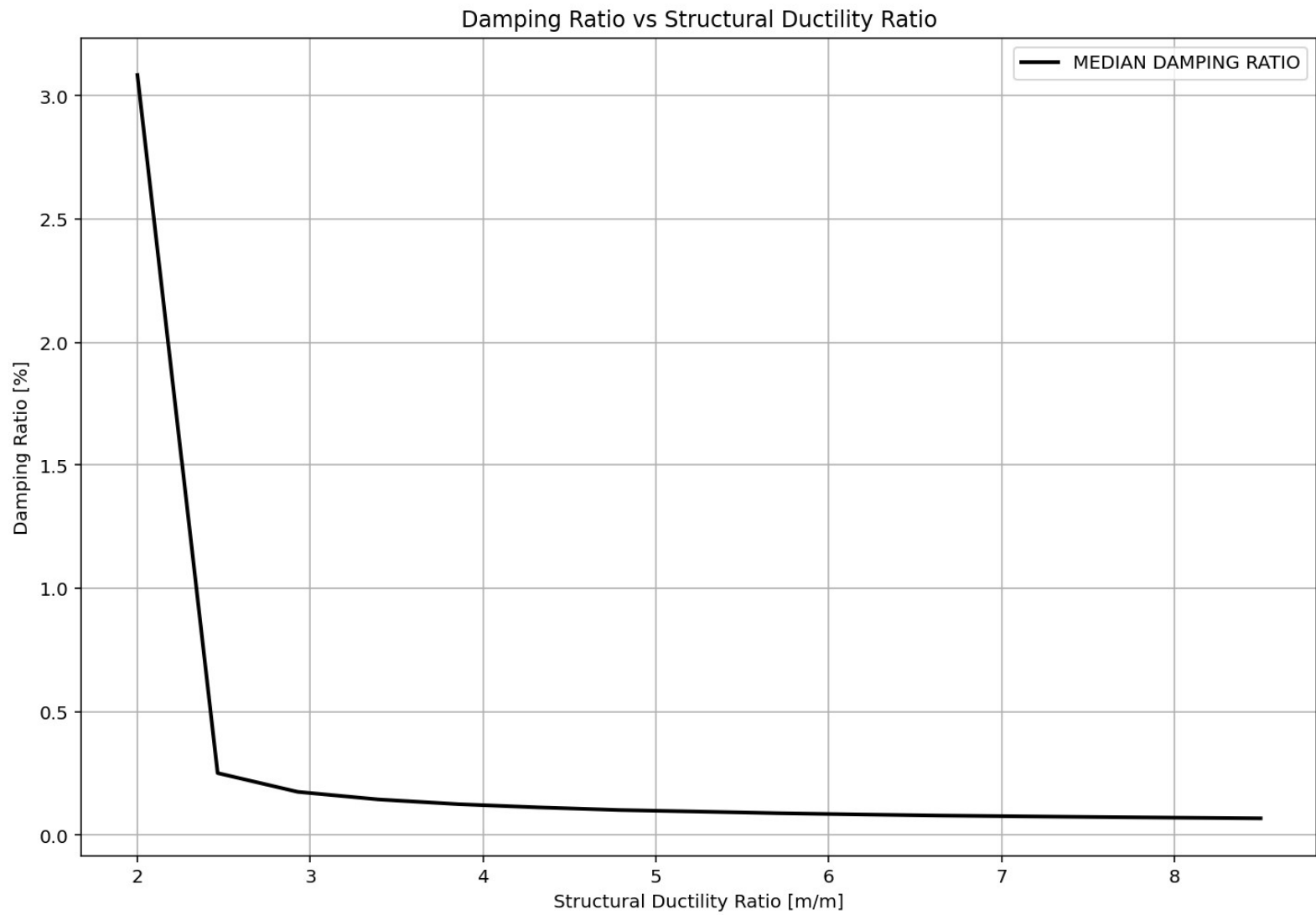


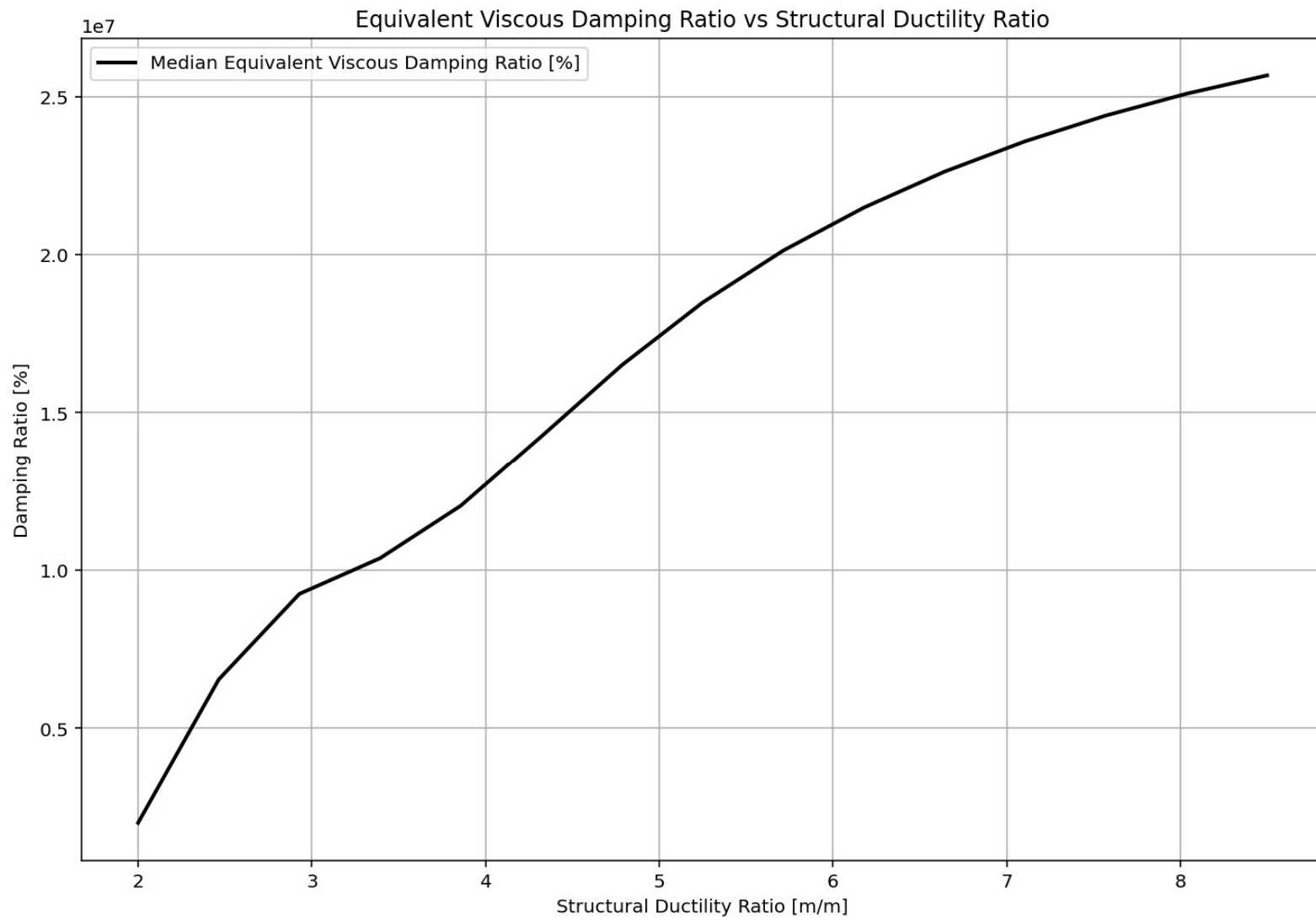




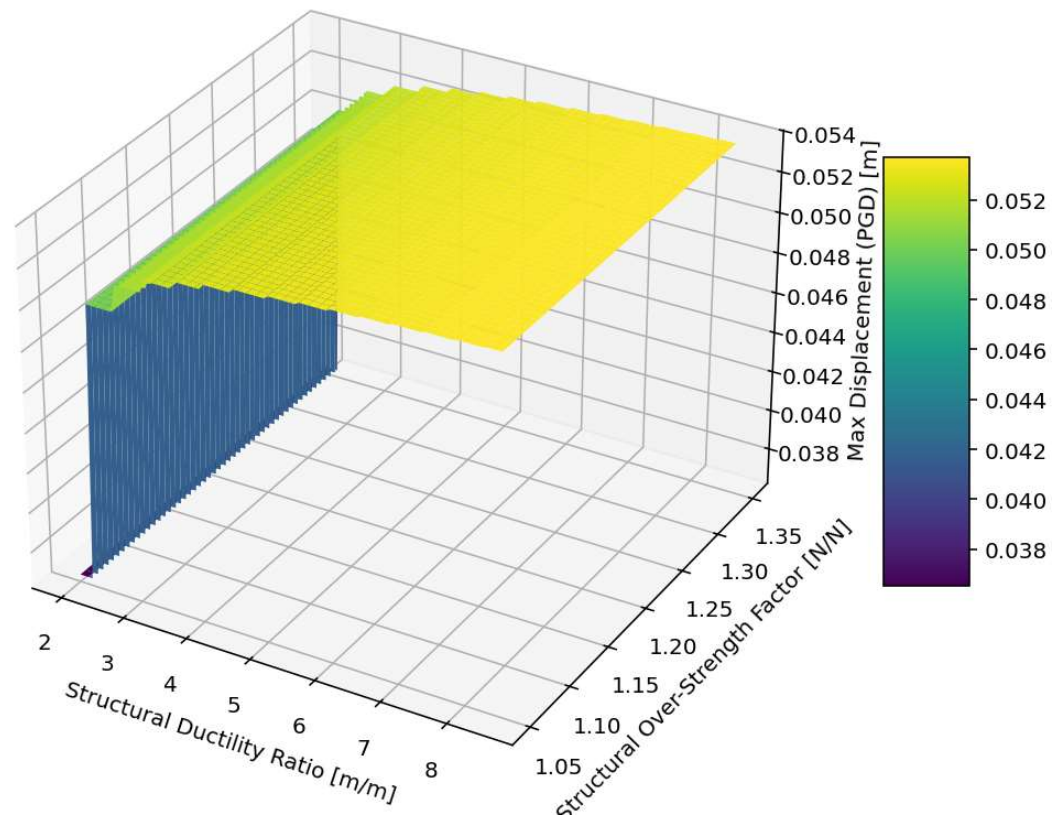




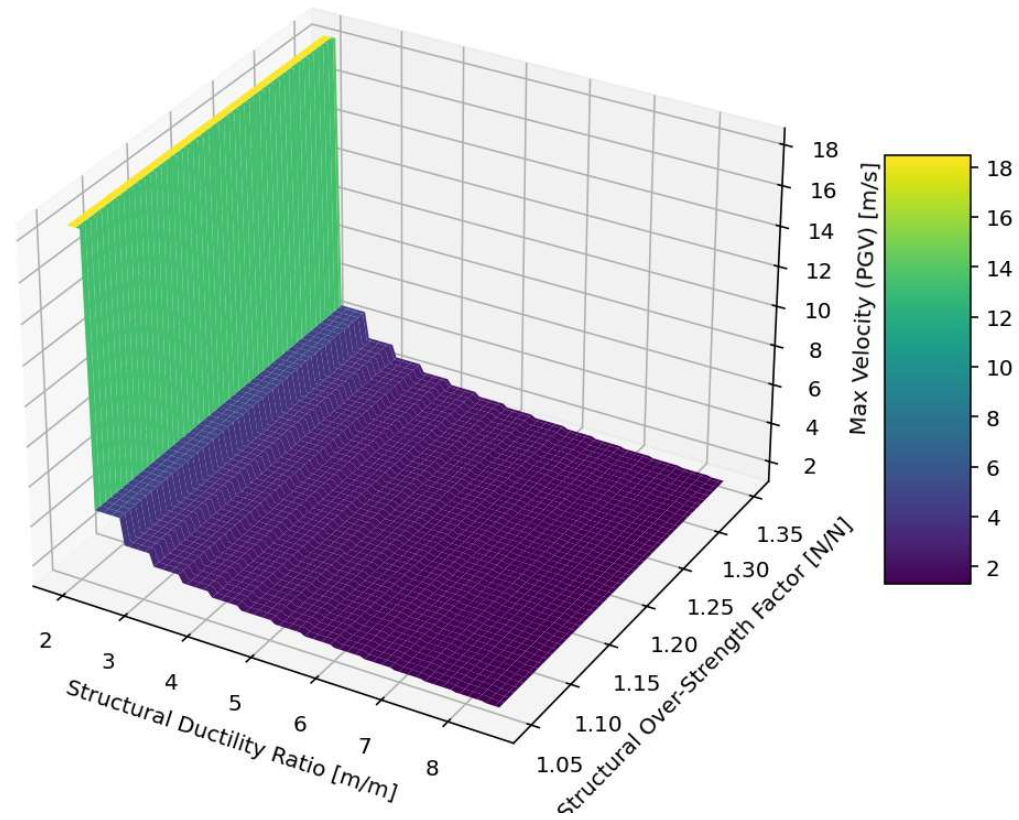




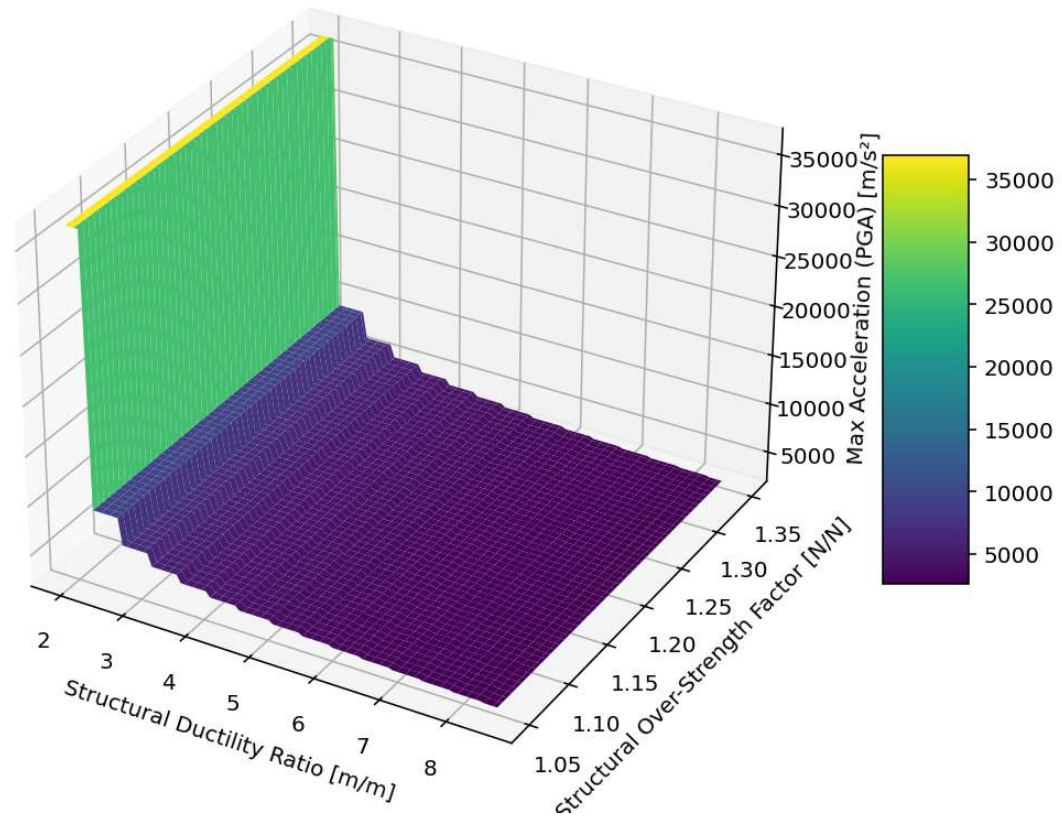
3D Contour Plot of Max Displacement (PGD) [m]



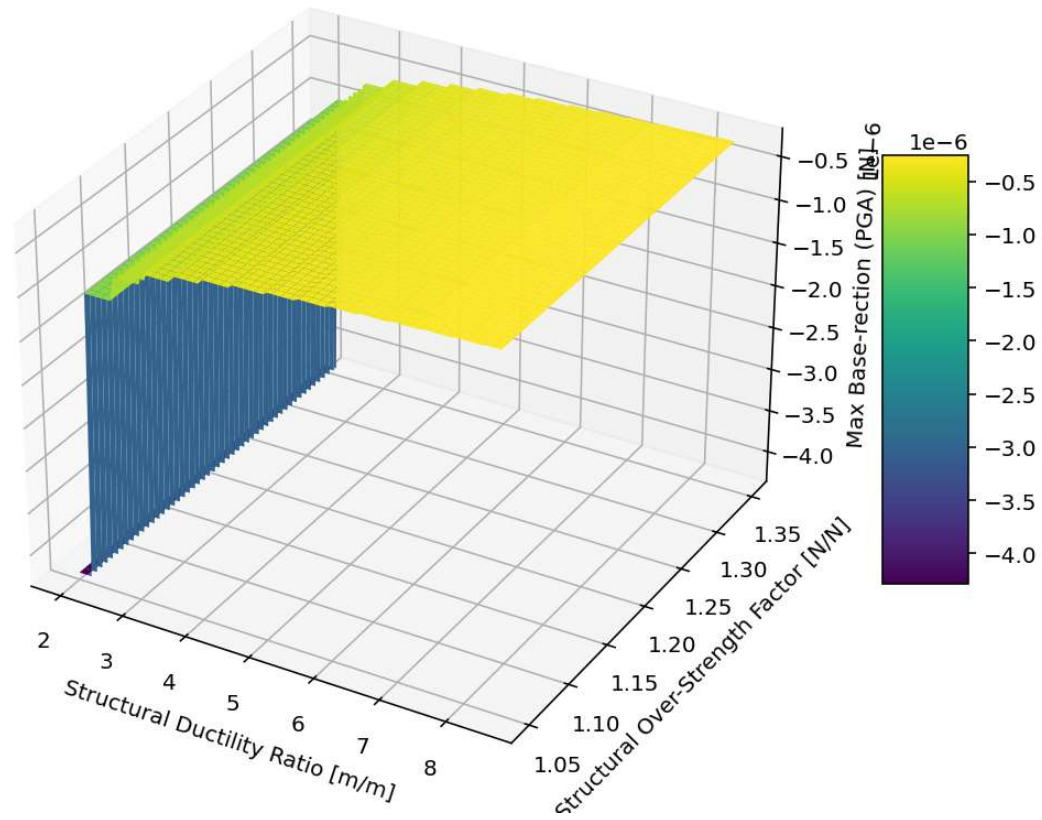
3D Contour Plot of Max Velocity (PGV) [m/s]



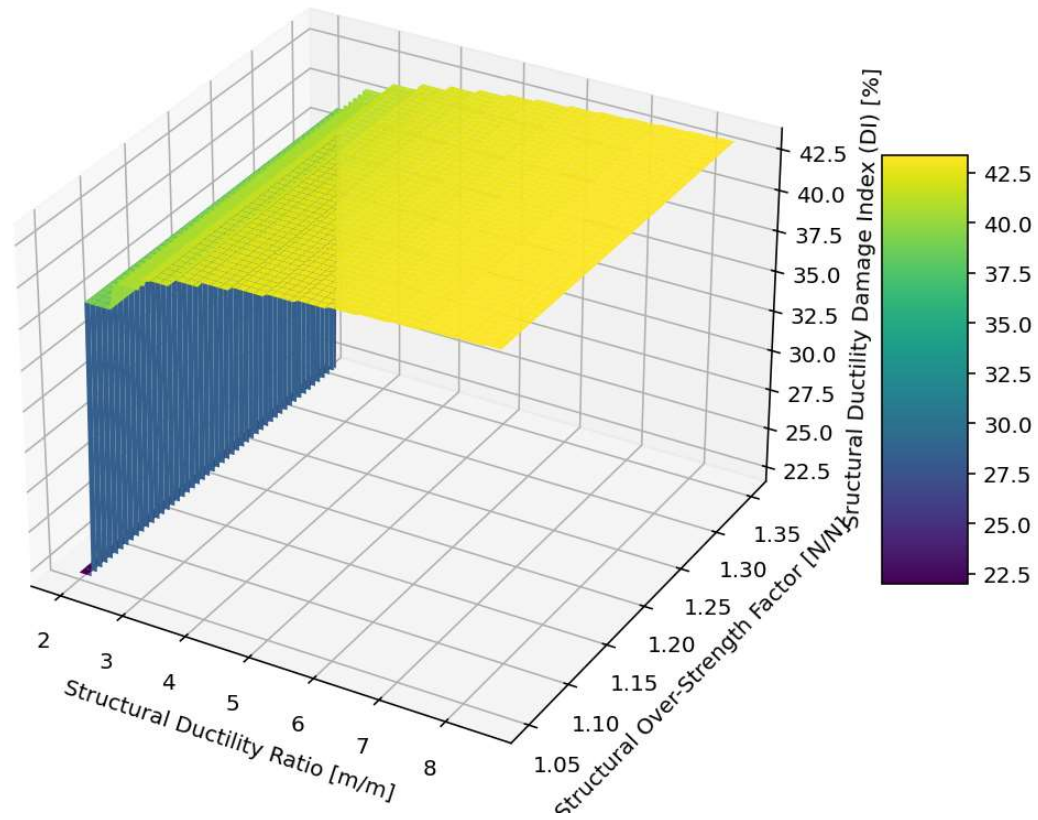
3D Contour Plot of Max Acceleration (PGA) [m/s^2]



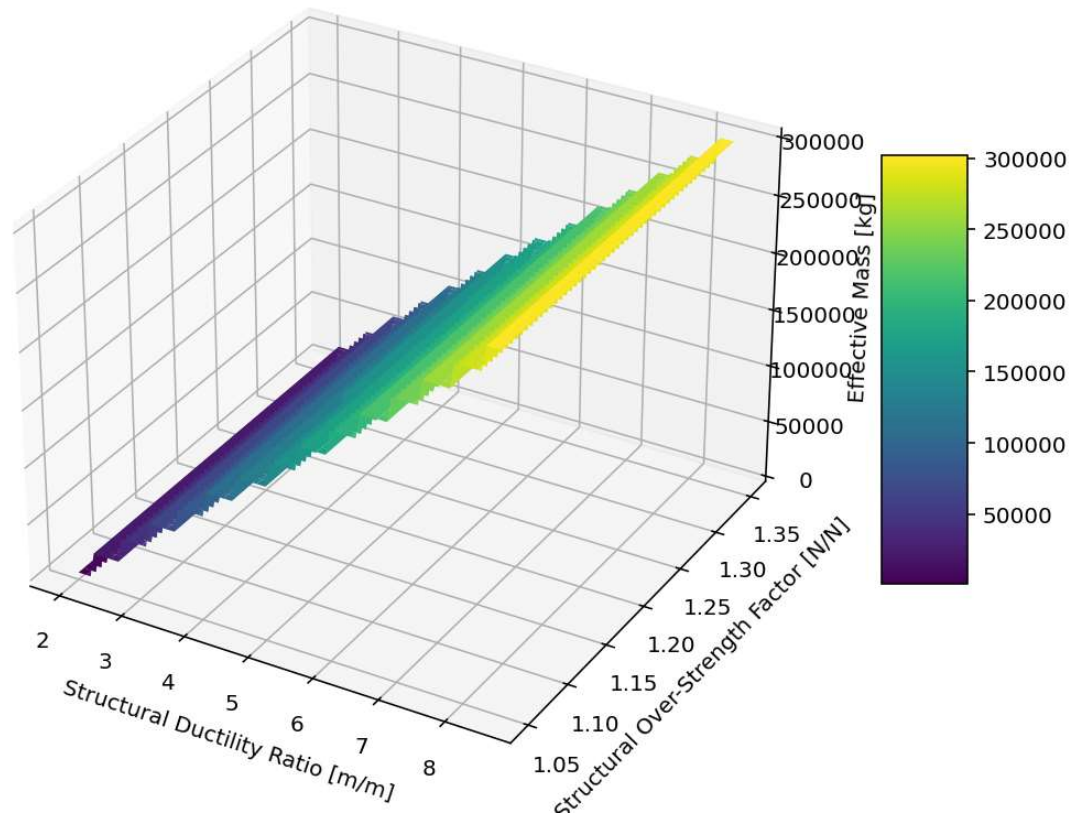
3D Contour Plot of Max Base-rection (PGA) [N]



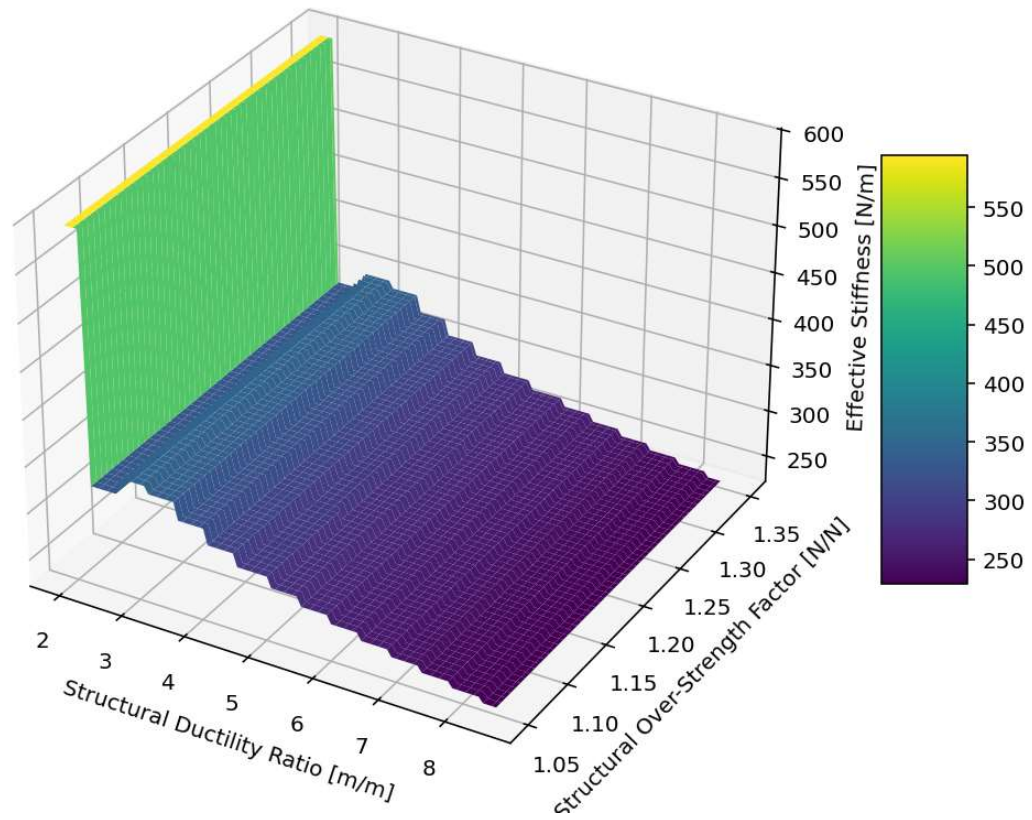
3D Contour Plot of Structural Ductility Damage Index (DI) [%]



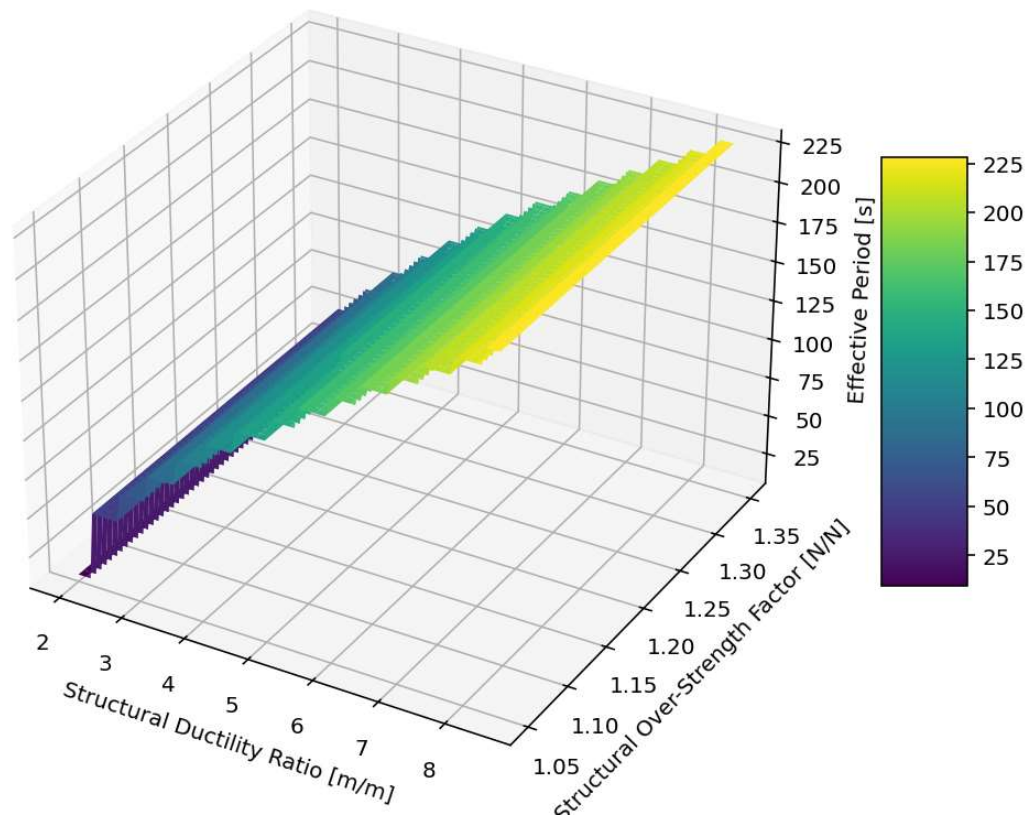
3D Contour Plot of Effective Mass [kg]



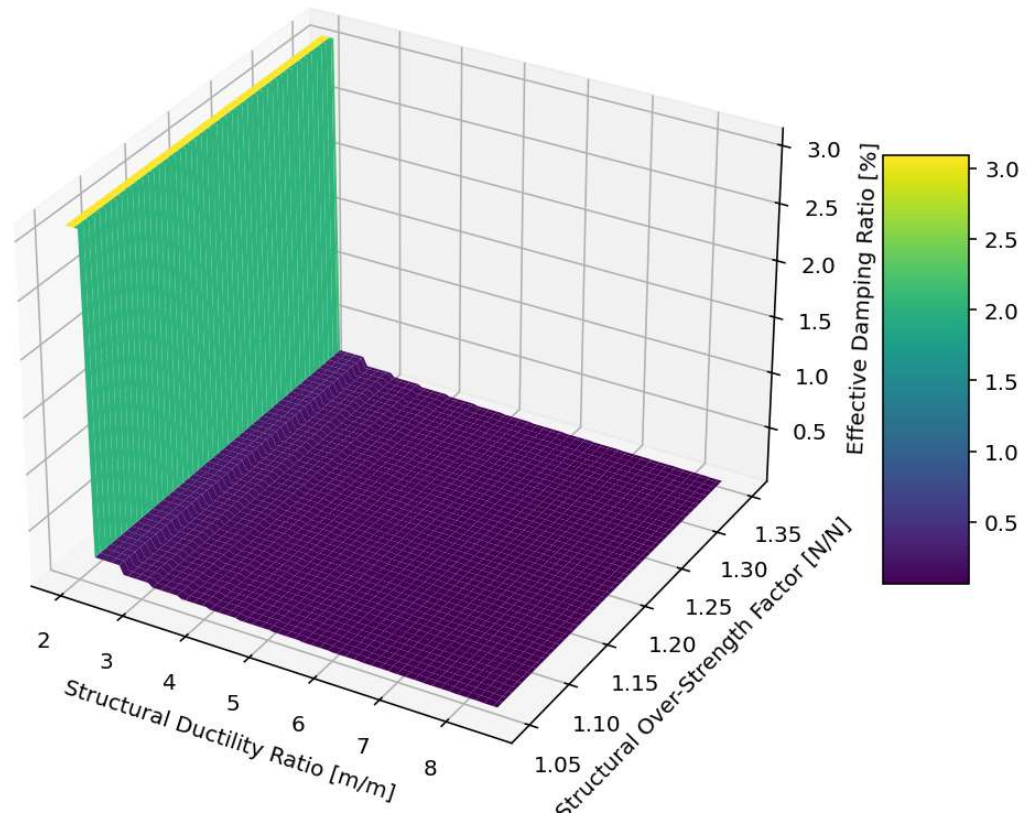
3D Contour Plot of Effective Stiffness [N/m]



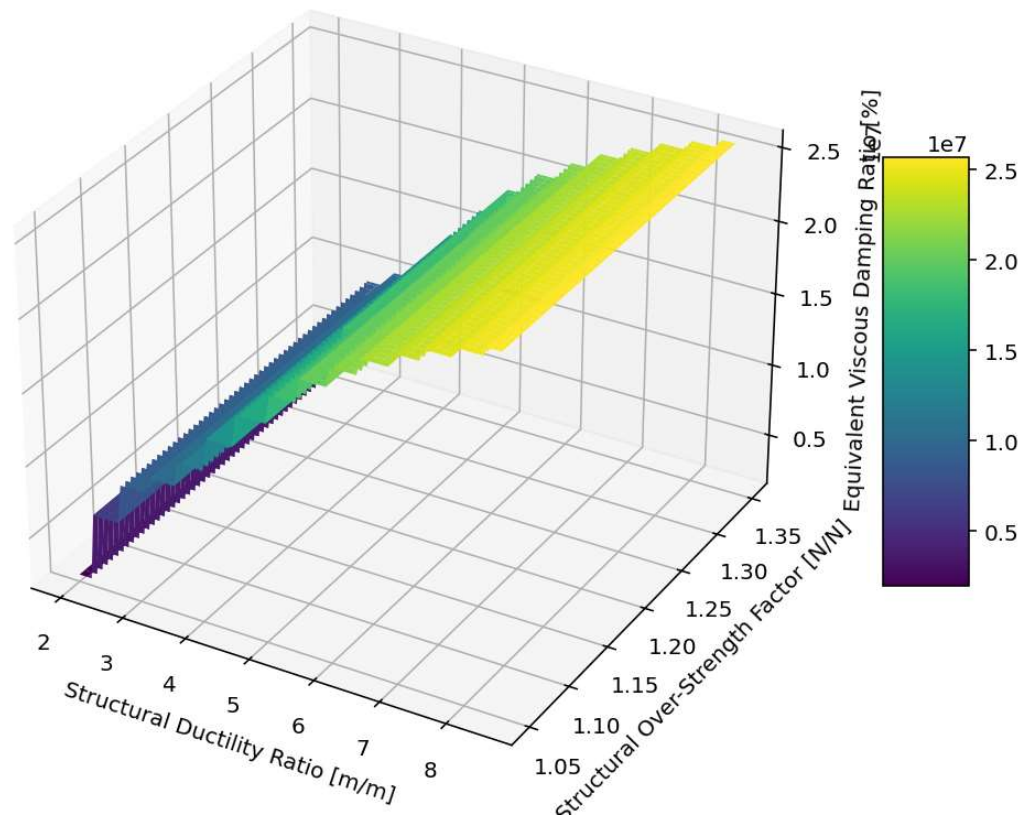
3D Contour Plot of Effective Period [s]



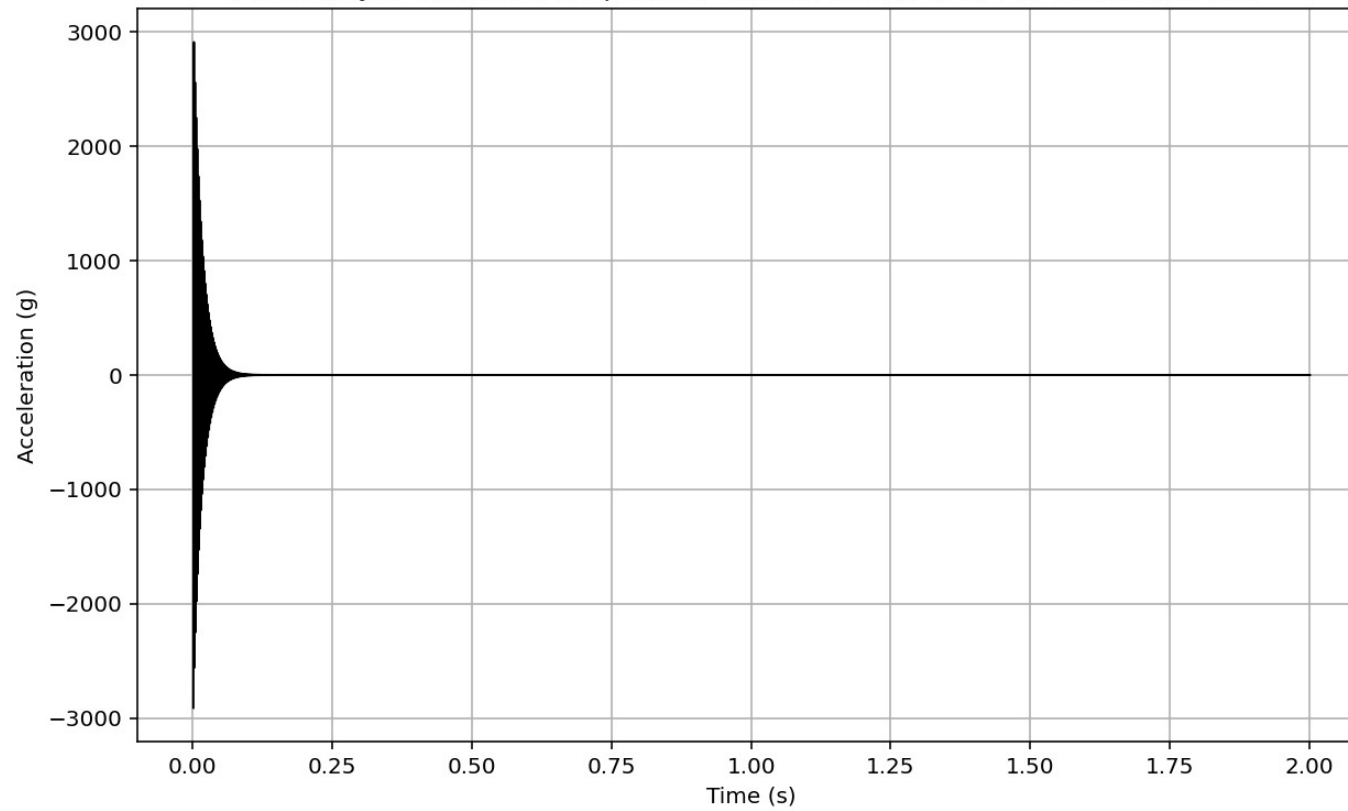
3D Contour Plot of Effective Damping Ratio [%]

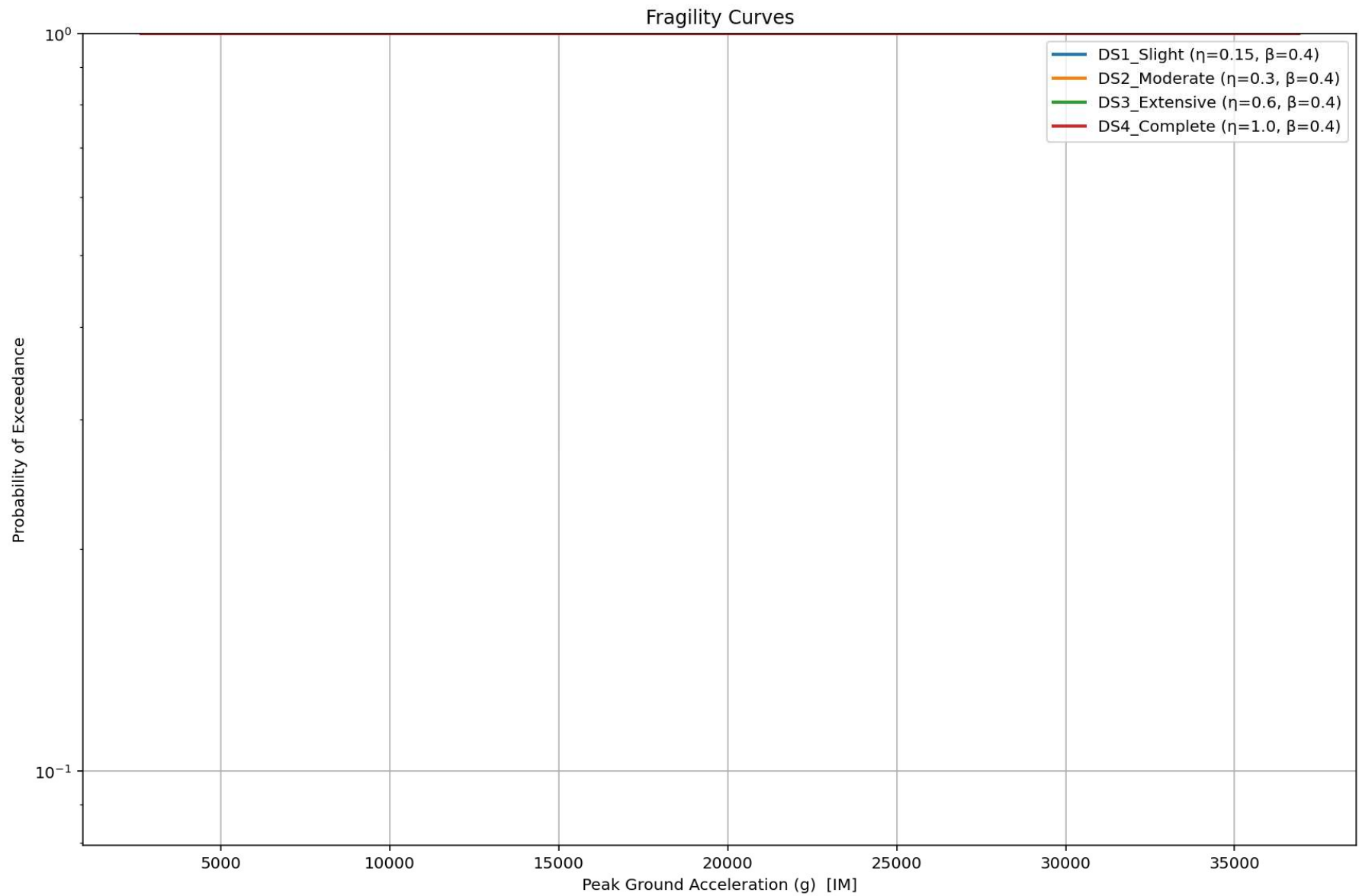


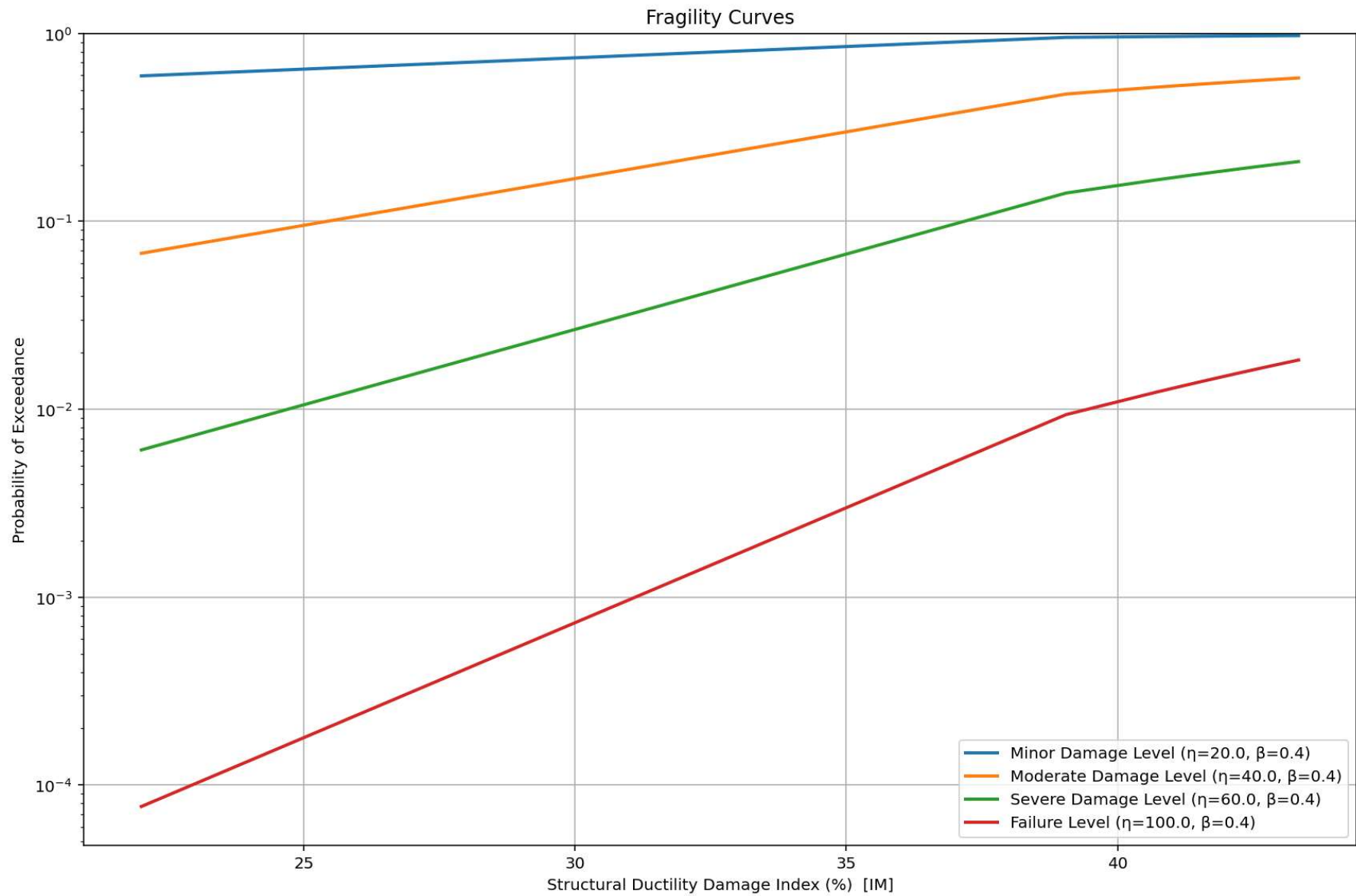
3D Contour Plot of Equivalent Viscous Damping Ratio [%]

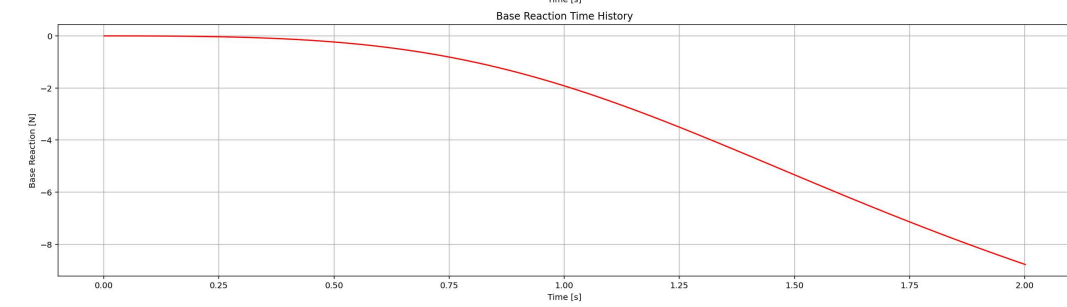
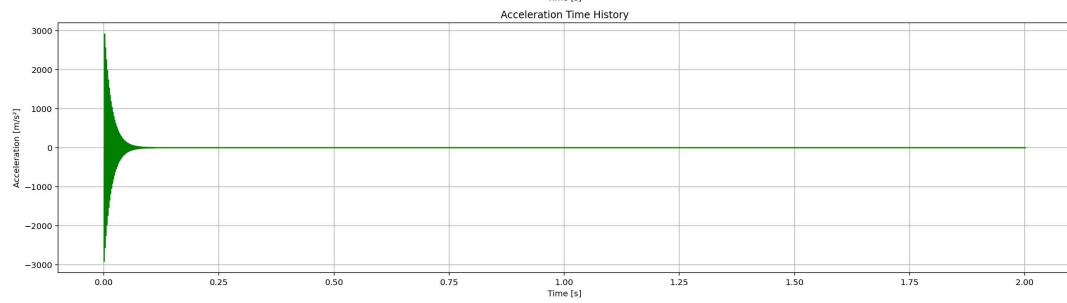
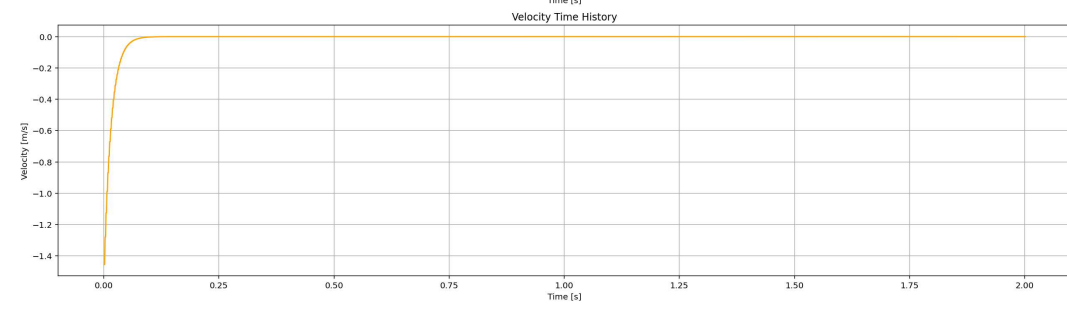
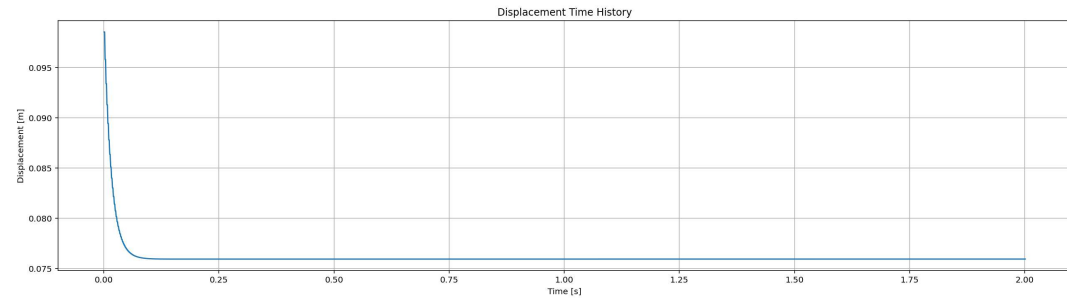


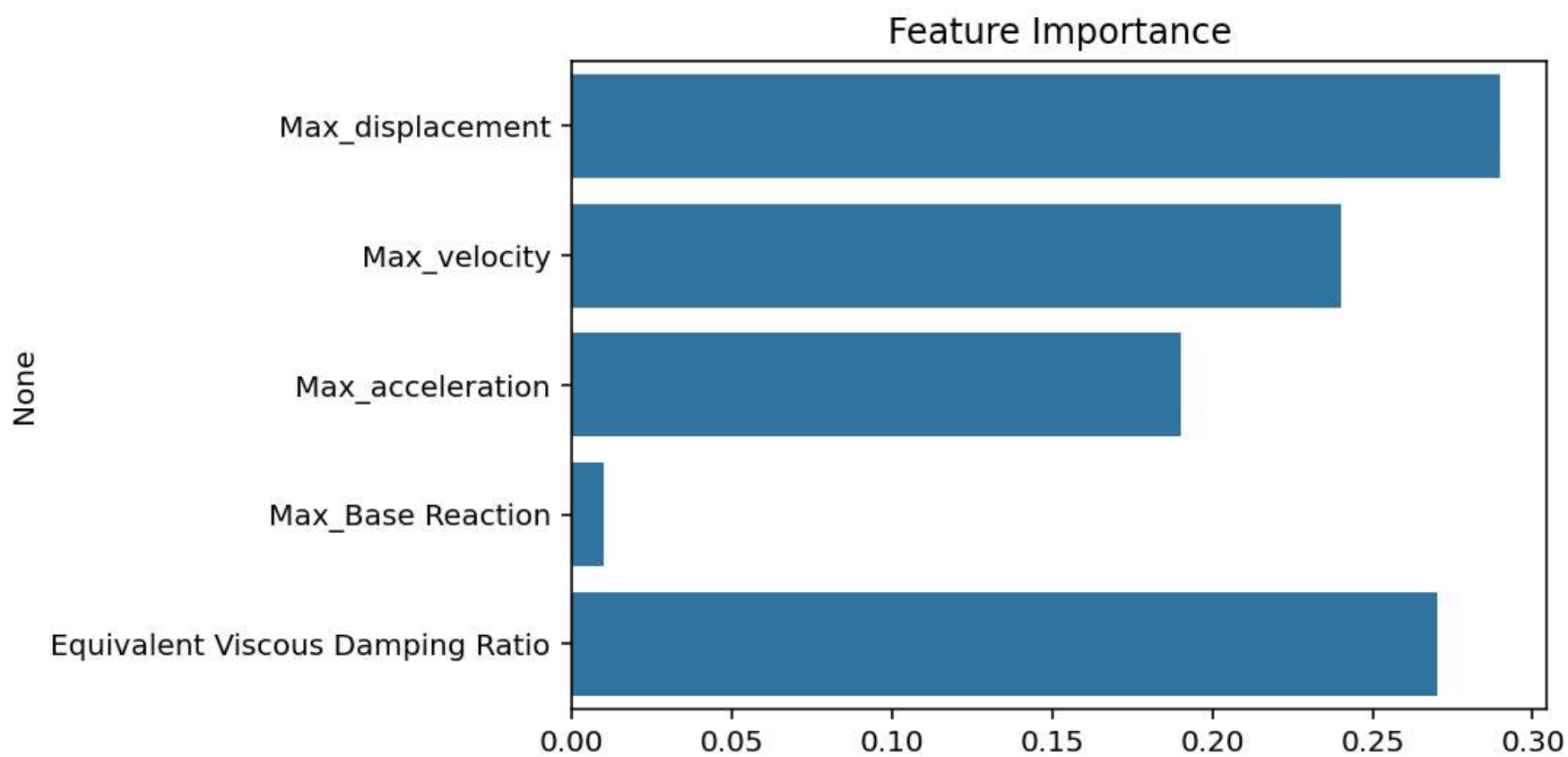
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 2911.4193











Correlation Heatmap

